

# Supplementary Information

Survival-supervised deconvolution identifies a replicable tumor–stroma prognostic architecture in pancreatic cancer

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## Part I

# Supplementary Methods

## 1 Model details

Let  $X \in \mathbb{R}_{\geq 0}^{p \times n}$  denote the nonnegative gene expression matrix with  $p$  genes and  $n$  subjects. DeSurv approximates  $X \approx WH$  with  $W \in \mathbb{R}_{\geq 0}^{p \times k}$  and  $H \in \mathbb{R}_{\geq 0}^{k \times n}$ , and links the shared program matrix  $W$  to survival via an elastic-net-penalized Cox model. Supervision is placed on the program matrix  $W$ , the basis that transfers to new cohorts, whereas the sample loadings  $H$  serve only to reconstruct the training expression matrix; new samples are therefore scored by projection onto the learned programs,  $Z = W^\top X$ , rather than by re-estimating loadings.

Let  $y \in \mathbb{R}_{> 0}^n$  be the observed survival times,  $\delta \in \{0, 1\}^n$  the event indicators, and  $Z = W^\top X \in \mathbb{R}^{k \times n}$  the sample-wise factor scores. We write  $Z_i$  for the  $i$ th column of  $Z$ , and  $n_{\text{event}} = \sum_{i=1}^n \delta_i$ . For convenience we recall the DeSurv loss from Equation 1 in the main text and rewrite it as

$$\begin{aligned} \mathcal{L}(W, H, \beta) = & (1 - \alpha) \left( \frac{1}{2\|X\|_F^2} \|X - WH\|_F^2 + \frac{\lambda_H}{2nk} \|H\|_F^2 \right) \\ & - \alpha \left( \frac{2}{n_{\text{event}}} \ell(W, \beta) - \lambda \{\xi \|\beta\|_1 + \frac{1-\xi}{2} \|\beta\|_2^2\} \right) \end{aligned}$$

where  $\|\cdot\|_F$  is the Frobenius norm and  $\beta \in \mathbb{R}^k$  are the Cox coefficients. The Cox log-partial likelihood  $\ell(W, \beta)$  is

$$\ell(W, \beta) = \sum_{i=1}^n \delta_i \left[ Z_i^\top \beta - \log \left( \sum_{j: y_j \geq y_i} \exp(Z_j^\top \beta) \right) \right]. \quad (1)$$

The inner sum runs over the risk set  $\{j : y_j \geq y_i\}$  of subjects still at risk at the observed time  $y_i$ ; tied event times are handled by the Breslow approximation in the `glmnet` Coxnet routine used for the  $\beta$  update (below). Hyperparameters satisfy  $\alpha \in [0, 1]$ ,  $\lambda_H \geq 0$ ,  $\lambda \geq 0$ , and  $\xi \in [0, 1]$ . The constants  $1/(2\|X\|_F^2)$ ,  $2/n_{\text{event}}$ , and  $1/(2nk)$  place the reconstruction, loadings-penalty, and survival terms on comparable numerical scales for stable optimization.

## 2 Optimization Algorithm

### 2.1 Block coordinate descent scheme

Optimization proceeds by block coordinate descent (BCD; Algorithm S1), which at each outer iteration updates  $H$ , then  $W$ , then  $\beta$  while holding the other blocks fixed.

### 2.2 Update for $H$

Conditional on  $W$ , the loss  $\mathcal{L}(W, H, \beta)$  reduces to a convex quadratic function of  $H$ . We adopt the standard NMF multiplicative update with  $\ell_2$  penalty:

$$H \leftarrow \max \left( H \odot \frac{W^\top X}{W^\top W H + \frac{\|X\|_F^2}{nk} \lambda_H H + \varepsilon_H}, \varepsilon_H \right), \quad (2)$$

---

**Algorithm S1** DeSurv block coordinate descent

---

**Input:**  $X \in \mathbb{R}_{\geq 0}^{p \times n}$ , survival times  $y \in \mathbb{R}_{> 0}^n$ , event indicators  $\delta \in \{0, 1\}^n$ , tolerance  $tol$ , maximum iterations  $maxit$ , supervision parameter  $\alpha$ , rank  $k$ , penalties  $\lambda_H$ ,  $\lambda$ , and  $\xi$

**Output:** Fitted DeSurv parameters  $(W, H, \beta)$

- 1: Initialize  $W^{(0)}, H^{(0)}$  with positive entries (e.g.,  $W^{(0)}, H^{(0)} \sim \text{Uniform}(0, \max X)$ )
  - 2: Initialize  $\beta^{(0)}$  (e.g.,  $\beta^{(0)} = \mathbf{1}$ )
  - 3:  $loss = \mathcal{L}(W^{(0)}, H^{(0)}, \beta^{(0)})$
  - 4:  $eps = \infty, t = 0$
  - 5: **while**  $eps \geq tol$  **and**  $t < maxit$  **do**
  - 6:   **(H-update)**
  - 7:     Update  $H^{(t+1)}$  from  $H^{(t)}$  using the multiplicative rule in Eq. 2,
  - 8:     holding  $W^{(t)}$  and  $\beta^{(t)}$  fixed.
  - 9:   **(W-update)**
  - 10:    Update  $W^{(t+1)}$  from  $W^{(t)}$  using the hybrid multiplicative rule in Eq. 6,
  - 11:    holding  $H^{(t+1)}$  and  $\beta^{(t)}$  fixed, and perform backtracking to ensure
  - 12:     $\mathcal{L}$  does not increase.
  - 13:   **( $\beta$ -update)**
  - 14:    Update  $\beta^{(t+1)}$  by solving the elastic-net Cox subproblem using cyclic coordinate descent to numerical tolerance
  - 15:    (Eq. 7), holding  $W^{(t+1)}$  and  $H^{(t+1)}$  fixed.
  - 16:     $lossNew = \mathcal{L}(W^{(t+1)}, H^{(t+1)}, \beta^{(t+1)})$
  - 17:     $eps = |lossNew - loss|/|loss|$
  - 18:     $loss = lossNew$
  - 19:     $t = t + 1$
  - 20: **end while**
  - 21: **return**  $\hat{W} = W^{(t)}, \hat{H} = H^{(t)}, \hat{\beta} = \beta^{(t)}$
- 

where  $\odot$  denotes elementwise multiplication, all divisions are elementwise, and  $\varepsilon_H > 0$  is a small floor to prevent entries from becoming exactly zero. This update is equivalent to a majorization–minimization step and guarantees a nonincreasing reconstruction term conditional on  $W^{1,2}$ . The  $\varepsilon_H$ -floor ensures that iterates remain in the interior of the nonnegative orthant, which keeps the multiplicative  $H$ -update numerically well-defined.

## 2.3 Update for $W$

For  $W$ , we construct a hybrid multiplicative update that balances the contributions of the NMF and Cox gradients. Let

$$\nabla_W \mathcal{L}_{\text{NMF}}(W, H) = \frac{1}{\|X\|_F^2} (WH - X)H^\top,$$

and let

$$\nabla_W \mathcal{L}_{\text{Cox}}(W, \beta) = \frac{2}{n_{\text{event}}} \nabla_W \ell(W, \beta).$$

where  $\nabla_W \ell(W, \beta)$  denotes the Cox gradient with respect to  $W$ . A closed-form expression for  $\nabla_W \ell$  is

$$\nabla_W \ell(W, \beta) = \sum_{i=1}^n \delta_i \left( x_i - \frac{\sum_{j: y_j \geq y_i} x_j \exp(x_j^\top W \beta)}{\sum_{j: y_j \geq y_i} \exp(x_j^\top W \beta)} \right) \beta^\top, \quad (3)$$

where  $x_i$  denotes the  $i$ th column of  $X$ .

We define a gradient-scaling heuristic

$$\zeta^{(t)} = \frac{\|(1 - \alpha) \nabla_W \mathcal{L}_{\text{NMF}}(W^{(t)}, H^{(t+1)})\|_F^2}{\|\alpha \nabla_W \mathcal{L}_{\text{Cox}}(W^{(t)}, \beta^{(t)})\|_F^2}, \quad (4)$$

and clip  $\zeta^{(t)}$  to avoid extreme values:  $\zeta^{(t)} \leftarrow \min(\zeta^{(t)}, \zeta_{\max})$  with  $\zeta_{\max} = 10^6$  in all experiments.

The multiplicative factor for  $W$  is then

$$R^{(t)} = \frac{\frac{(1-\alpha)}{\|X\|_F^2} X H^{(t+1)\top} + \frac{2\alpha}{n_{\text{event}}} \zeta^{(t)} \nabla_W \ell(W^{(t)}, \beta^{(t)})}{\frac{(1-\alpha)}{\|X\|_F^2} W^{(t)} H^{(t+1)} H^{(t+1)\top} + \varepsilon_W}, \quad (5)$$

and the proposed update is

$$W^{(t+1)} = \max(W^{(t)} \odot R^{(t)}, \varepsilon_W), \quad (6)$$

with a small floor  $\varepsilon_W > 0$ . When  $\alpha = 0$ , this reduces to the standard multiplicative update for NMF<sup>1</sup>; for  $\alpha > 0$ , the Cox gradient perturbs the update in a direction that decreases the supervised loss.

Because the combined multiplicative step does not, by itself, guarantee a nonincrease in the full loss  $\mathcal{L}$ , we embed it in a backtracking line search.

---

**Algorithm S2**  $W$  update with backtracking

---

**Input:** Value of  $W$  and the previous iteration  $t$ :  $W^{(t)}$ , the multiplicative update term at the current iteration  $R^{(t)}$ , the max number of backtracking updates  $max\_bt$ , the backtracking parameter  $\rho \in (0, 1)$

**Output:**  $W^{(t+1)}$

```

1:  $\theta = 1$ 
2:  $b = 1$ 
3:  $flag\_accept = FALSE$ 
4: while  $b \leq max\_bt$  do
5:    $W_{cand}^{(t+1)} = \max(W^{(t)} \odot [(1 - \theta) + \theta R^{(t)}], \varepsilon_W)$ 
6:   (Column normalization)
7:     Compute  $D = \text{diag}(\|W_{(:,1)cand}^{(t+1)}\|_2, \dots, \|W_{(:,k)cand}^{(t+1)}\|_2)$ 
8:     and set  $W_{cand}^{(t+1)} \leftarrow W_{cand}^{(t+1)} D^{-1}$ ,  $H_{cand}^{(t+1)} \leftarrow D H^{(t+1)}$ , and  $\beta_{cand}^{(t)} \leftarrow D \beta^{(t)}$ 
9:     if  $\mathcal{L}(W_{cand}^{(t+1)}, H_{cand}^{(t+1)}, \beta_{cand}^{(t)}) \leq \mathcal{L}(W^{(t)}, H^{(t+1)}, \beta^{(t)})$  then
10:       $W^{(t+1)} = W_{cand}^{(t+1)}$ 
11:       $H^{(t+1)} = H_{cand}^{(t+1)}$ 
12:       $\beta^{(t)} = \beta_{cand}^{(t)}$ 
13:       $flag\_accept = TRUE$ 
14:      break
15:   end if
16:    $\theta = \theta * \rho$ 
17:    $b = b + 1$ 
18: end while
19: if  $flag\_accept = FALSE$  then
20:    $W^{(t+1)} = W^{(t)}$ 
21: end if
```

---

Column normalization preserves the reconstruction  $WH$  and linear predictor  $W\beta$ . Because the accompanying rescaling can alter penalization terms, the normalized proposal is evaluated using the full penalized objective within the backtracking procedure, which accepts the update only if it does not increase  $\mathcal{L}$ .

## 2.4 Update for $\beta$

Conditional on  $(W, H)$ , the loss in  $\beta$  reduces to a convex elastic-net-penalized Cox problem; the common factor  $\alpha > 0$  multiplies both the log-partial-likelihood and the penalty in the joint loss (Equation 1) and is

dropped here, as it does not affect the minimizer:

$$\min_{\beta \in \mathbb{R}^k} \left\{ -\frac{2}{n_{\text{event}}} \ell(W, \beta) + \lambda \left( \xi \|\beta\|_1 + \frac{1-\xi}{2} \|\beta\|_2^2 \right) \right\}.$$

We solve this subproblem by cyclic coordinate descent following<sup>3</sup>. Writing  $\ell(\beta) = \ell(W, \beta)$  and  $\tilde{\eta}_i = Z_i^\top \beta$ , the update for coordinate  $r$  has the closed form

$$\hat{\beta}_r = \frac{S\left(\frac{1}{n} \sum_{i=1}^n w(\tilde{\eta})_i z_{i,r} \left[ v(\tilde{\eta})_i - \sum_{j \neq r} z_{i,j} \beta_j \right], \lambda \xi\right)}{\frac{1}{n} \sum_{i=1}^n w(\tilde{\eta})_i z_{i,r}^2 + \lambda(1-\xi)}, \quad (7)$$

where  $S(a, \tau) = \text{sign}(a) \max(|a| - \tau, 0)$  is the soft-thresholding operator, and  $w(\tilde{\eta})$ ,  $v(\tilde{\eta})$  are standard local quadratic approximations of the Cox log-partial likelihood<sup>3</sup>. We iterate coordinate updates until the subproblem converges to numerical tolerance, yielding a minimizer in  $\beta$  for the current  $W$ .

## 2.5 Normalization of $W$

After each accepted  $W$  update, we normalize the columns of  $W$  as  $W \leftarrow WD^{-1}$ , and adjust  $H$  and  $\beta$  as

$$H \leftarrow DH, \quad \beta \leftarrow D\beta,$$

where  $D$  is the diagonal matrix of column  $\ell_2$ -norms of  $W$ . This preserves the reconstruction  $WH$  and the linear predictor  $W\beta$ ; because the accompanying rescaling of  $H$  and  $\beta$  can alter the penalization terms, the normalized proposal is evaluated under the full penalized objective within the backtracking procedure for the  $W$  update rather than assumed loss-invariant. Normalization prevents degeneracy in the scale-nonidentifiable factorization and keeps columns of  $W$  comparable in magnitude and interpretable as gene programs.

## 3 Supervision on $W$ versus $H$

Classical nonnegative matrix factorization (NMF) is non-identifiable: for any invertible matrix  $R$  with  $WR \geq 0$  and  $R^{-1}H \geq 0$ , the factorization  $(W, H)$  can be transformed to  $(WR, R^{-1}H)$  without changing the reconstruction error, yielding multiple valid solutions<sup>4-6</sup>. Although normalizing columns of  $W$  or rows of  $H$  resolves the trivial scaling ambiguity, nonnegative mixing transformations persist, and empirical studies consistently show that the sample-loading matrix  $H$  is more sensitive to initialization and local minima than the program matrix  $W$ <sup>7-9</sup>.

In DeSurv, survival supervision enters through the projection  $Z = W^\top X$  in the partial log-likelihood, so that the gradient of the DeSurv loss with respect to the programs is

$$\nabla \mathcal{L} = (1 - \alpha) \nabla_W \mathcal{L}_{\text{NMF}}(W, H) - \alpha \nabla_W \mathcal{L}_{\text{Cox}}(W, \beta),$$

and therefore the update rule for  $W$  as in Equation 6 relies on both the reconstruction term and the supervision term. As such, the gene-program definitions are directly shaped by their association with survival. This mechanism amplifies genes aligned with the hazard gradient and suppresses those that dilute prognostic structure, favoring programs enriched for survival-relevant variation.

By contrast, if supervision were applied to the sample loadings  $H$ , the model could reduce the Cox loss by redistributing patient-specific coefficients while leaving the gene-level programs  $W$  nearly unchanged, a behavior documented in multiple supervised variants of NMF and supervised topic models, where supervision applied only to the coefficient matrix modifies  $H$  but not the basis  $W$ <sup>10,11</sup>.

Supervising  $W$  also provides a practical advantage for **portability**: because  $W$  defines gene-level programs, new samples can be embedded by the closed-form projection  $Z_{\text{new}} = W^\top X_{\text{new}}$ . In contrast, supervising  $H$  would not yield such a mapping; instead, obtaining sample loadings for external datasets would require

solving a nonnegative least-squares (NNLS) problem for each new sample, introducing optimization noise and reducing reproducibility.

Together, these considerations motivate supervising through  $W$ , which shapes the gene programs toward prognostic directions, stabilizes solutions across initializations, and yields biologically interpretable signatures that generalize across cohorts.

## 4 Cross-validation procedure

Algorithm S3 describes the cross validation procedure with  $F$  folds and  $R$  initializations. We define the c-index using comparable pairs  $\mathcal{P} = \{(i, j) : y_i < y_j, \delta_i = 1\}$  and linear predictors  $\hat{\eta}_i$ :

$$\hat{c} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} [\mathbb{1}(\hat{\eta}_i > \hat{\eta}_j) + \frac{1}{2} \mathbb{1}(\hat{\eta}_i = \hat{\eta}_j)]. \quad (8)$$

---

### Algorithm S3 Cross-validation for DeSurv pipeline

---

**Input:**

- Number of folds  $F$
- Number of initializations  $R$
- Observed data  $(X, y, \delta)$
- Hyperparameters  $k, \alpha, \lambda, \xi, \lambda_H$ , and number of top genes per factor  $n_{top}$

**Output:** The cross-validated c-index

- 1: Divide subjects into  $F$  folds.
  - 2: **for**  $f = 1$  to  $F$  **do**
  - 3:     Split data into training and validation  $(X_{(-f)}, y_{(-f)}, \delta_{(-f)})$ , and  $(X_{(f)}, y_{(f)}, \delta_{(f)})$  sets
  - 4:     **for**  $r = 1$  to  $R$  **do**
  - 5:         set.seed( $r$ )
  - 6:         Apply Algorithm S1 with inputs  $(X_{(-f)}, y_{(-f)}, \delta_{(-f)})$ , and  $(k, \alpha, \lambda, \xi, \lambda_H)$
  - 7:         Obtain  $\hat{W}$  and  $\hat{\beta}$  as output from Algorithm S1
  - 8:         Restrict  $\hat{W}$  to the top  $n_{top}$  genes per factor (by factor-specificity score) and compute the estimated linear predictor on this support:  $\hat{\eta} = X_{(f)}^T \hat{W} \hat{\beta}$
  - 9:         Compute c-index, denoted  $\hat{c}_{(f)r}$ , according to Equation 8
  - 10:     **end for**
  - 11:     **end loop over**  $r$
  - 12:     Compute average c-index across initializations:  $\hat{c}_{(f)} = \frac{1}{R} \sum_{r=1}^R \hat{c}_{(f)r}$
  - 13: **end for**
  - 14: **end loop over**  $f$
  - 15: Compute final cross-validated c-index:  $\hat{c} = \frac{1}{F} \sum_{f=1}^F \hat{c}_{(f)}$
  - 16: **return** Final estimate  $\hat{c}$
- 

## 5 Bayesian Optimization

We treat hyperparameter tuning as a black-box optimization problem over a bounded domain. Let  $\theta$  collect the tuned parameters (at minimum  $\theta = (k, \alpha, \lambda, \xi)$ , and optionally discrete choices such as the signature size  $n_{top}$ ). For a given  $\theta$  we run the cross-validation procedure in Algorithm S3, averaging the c-index across folds and random initializations to form the objective

$$\hat{c}(\theta) = \frac{1}{F} \sum_{f=1}^F \hat{c}_{(f)}(\theta).$$

Because  $\hat{c}(\theta)$  is expensive to evaluate and non-differentiable with respect to  $\theta$ , we use Bayesian optimization (BO) to adaptively select candidate configurations.

We place a Gaussian-process (GP) prior on the response surface  $f(\theta) \sim \mathcal{GP}(0, k(\theta, \theta'))$  with a Mat'ern kernel and automatic relevance determination length-scales. Continuous parameters are rescaled to  $[0, 1]$  and penalties are optimized on a log scale; integer-valued coordinates (e.g.,  $k$  or  $n_{\text{top}}$ ) are proposed in the continuous space and rounded inside the evaluator. After an initial batch of evaluations, the GP is refit after each new observation and used to guide acquisition.

At iteration  $t$ , let  $\mu_t(\theta)$  and  $\sigma_t(\theta)$  denote the GP posterior mean and standard deviation, and let  $f_\star$  be the best observed score. We select new candidates by maximizing expected improvement (EI),

$$\text{EI}(\theta) = (\mu_t(\theta) - f_\star - \kappa) \Phi(Z(\theta)) + \sigma_t(\theta) \phi(Z(\theta)), \quad Z(\theta) = \frac{\mu_t(\theta) - f_\star - \kappa}{\sigma_t(\theta)},$$

where  $\Phi$  and  $\phi$  are the standard normal cdf/pdf and  $\kappa \geq 0$  controls the exploration margin. EI is evaluated over a candidate pool sampled from the bounded search region, and the maximizer is evaluated next. The loop continues until the iteration budget is exhausted or the improvement plateaus.

For the PDAC application, the objective was the mean 5-fold cross-validated Harrell C-index (Algorithm S3), with folds stratified by cohort under a fixed seed (123); each configuration was evaluated from 30 random initializations and its C-index averaged across them. The bounded search domain was  $k \in [2, 12]$ ,  $\alpha \in [0, 1]$ ,  $\lambda \in [10^{-3}, 10^3]$  (optimized on a  $\log_{10}$  scale),  $\xi \in [0, 1]$ , and  $n_{\text{top}} \in [50, 300]$ , with the loadings ridge penalty fixed at  $\lambda_H = 0$ . Bayesian optimization used an initial design of 50 evaluations followed by 100 GP-guided iterations (150 total evaluations), maximizing expected improvement over a candidate pool of 4,000 points sampled from the bounded domain at each step with an exploration margin  $\kappa = 0.01$ . The unsupervised comparator fixed  $\alpha = 0$  and tuned the remaining coordinates identically.

For model selection, we take the highest observed  $\hat{c}(\theta)$  as the optimal configuration. When fold-level standard errors are available, we apply a one-standard-error rule for  $k$ , selecting the smallest rank whose mean performance lies within one standard error of the best. The final DeSurv model is refit at the selected parameters using the consensus-based initialization described above, and the BO-chosen  $n_{\text{top}}$  (when tuned) is used to truncate each program for downstream analysis.

The parameter  $n_{\text{top}}$  controls which genes enter the survival linear predictor. After factorization, each sample's score for factor  $j$  is computed as  $\widehat{W}_j^\top x$ , where  $\widehat{W}_j$  retains only the  $n_{\text{top}}$  genes with the largest factor-specificity scores  $s_{ij}$  (Supplementary Information, Section 8); genes outside this set still shape the factorization but do not enter the linear predictor. BO selects  $n_{\text{top}}$  jointly with  $k$ ,  $\alpha$ ,  $\lambda$ , and  $\xi$  by optimizing cross-validated C-index. When  $\alpha = 0$  (no survival supervision), gene selection and survival outcomes are decoupled.

Supplementary Fig. S1 illustrates the tuned response surface for the two most influential coordinates in the PDAC training cohort: the cross-validated C-index over factorization rank  $k$  and supervision strength  $\alpha$ . Performance is maximized at intermediate supervision and moderate rank and degrades toward the unsupervised corner ( $\alpha = 0$ ), consistent with a benefit of survival supervision; this is an empirical model-selection sensitivity surface, not a simulation result.

## 5.1 Consensus initialization for the final model

The final DeSurv model at the selected hyperparameters was initialized from a consensus of multiple independent runs to reduce sensitivity to random starts. We first fit  $R = 100$  DeSurv models to the training data from independent random nonnegative initializations at the selected  $(k, \alpha, \lambda, \xi, n_{\text{top}})$  (seed 1–100). For each run, the top  $n_{\text{top}}$  genes of every learned program were extracted by factor-specificity score, and a gene-by-gene consensus matrix was accumulated across runs, recording how often each pair of genes appeared together among the top genes of the same program. Genes appearing in the top sets of fewer than  $\lceil 0.3R \rceil$  of the  $R$  runs were treated as inactive and excluded from clustering. The consensus matrix over the remaining genes was partitioned into  $k$  groups by agglomerative hierarchical clustering (average linkage), and each cluster defined one column of the initial gene-program matrix  $W_0$ ; inactive genes and otherwise-empty entries

were set to a small positive floor ( $\sqrt{\epsilon_{\text{mach}}}$ ) to preserve strict positivity. The initial loadings  $H_0$  were obtained by nonnegative least squares given  $W_0$  and the training expression matrix, and the initial Cox coefficients  $\beta_0$  were set to zero. This consensus seed used only training-cohort expression and was applied solely to initialize the single final refit after hyperparameter selection; it used no validation-cohort data and therefore introduced no information leakage.

## 6 PDAC Datasets

We analyzed PDAC cohorts from TCGA-PAAD and CPTAC-PDAC as training data<sup>12,13</sup>. External validation used independent cohorts from Dijk, Moffitt, PACA-AU (array and RNA-seq), and Puleo<sup>14–17</sup>. Training models were fit on the combined TCGA and CPTAC expression matrices after harmonizing gene identifiers; validation datasets were processed with the same gene set and transformations before projection, scoring and downstream validation analyses.

Dataset-specific inclusion and preprocessing steps were as follows. For TCGA-PAAD, we excluded samples lacking tumor grade and mapped features to gene symbols. For CPTAC, we retained samples annotated as PDAC by histology. For Dijk, all 90 samples passed quality-control filters and were retained. For Moffitt, only primary tumor specimens were kept. For PACA-AU array and RNA-seq, we retained primary PDAC tumors and collapsed duplicated gene symbols by median; the RNA-seq cohort received a “\_seq” suffix on sample IDs to avoid collisions with the array cohort. Puleo required no additional dataset-specific filtering beyond survival QC.

Across cohorts, samples without valid survival outcomes (nonpositive time, missing time, or missing event indicator) were removed. Expression matrices were analyzed on a log scale: RNA-seq cohorts as  $\log_2(\text{TPM} + 1)$ ; microarray cohorts in platform-normalized log-scale intensities. For each training cohort (TCGA-PAAD and CPTAC) separately, genes were ranked by mean expression and by variance independently; the 3,000 genes with the lowest sum of these two ranks were retained, selecting genes that are both highly expressed and highly variable. The intersection of these two gene sets yielded 1,970 genes used for model training. Validation datasets were restricted to this same 1,970-gene set. A within-sample rank transformation was then applied to every dataset to mitigate platform and scale effects. Specifically, for each sample the expression values across the 1,970 retained genes were replaced by their ascending ranks (ties resolved by average ranks), so that every sample carries the same nonnegative rank distribution over the shared gene set; ranks are inherently nonnegative and therefore preserve NMF compatibility. Validation cohorts were rank-transformed by exactly the same per-sample procedure before projection, with no use of validation outcomes.

Table S1 summarizes the cohorts, platforms, sample sizes after quality control, and patient-level metadata used in this study. Overall survival time and event indicators were available for all cohorts. PurIST<sup>18</sup> and DeCAF<sup>19</sup> molecular classifications, applied to bulk expression data as described in those references, were used as covariates in the covariate-adjusted external validation analysis. Gene expression data and molecular classifications for the validation cohorts were originally compiled for the DeCAF study<sup>19</sup>; training data were obtained directly from GDC.

## 7 Survival Analysis

Survival analyses used overall survival time with event indicators as provided by each cohort. For fitted DeSurv models, sample-level program scores were computed as  $Z = \bar{W}^\top X$  and standardized using the training mean and standard deviation before constructing the linear predictor  $\hat{\eta} = Z\hat{\beta}$ . For the per-factor associations shown in the validation forest plot, each individual factor score was standardized to zero mean and unit standard deviation within its own cohort, so that per-factor hazard ratios are expressed per one standard-deviation increment; the multivariable linear predictor was likewise reported per standard deviation. Prognostic performance was summarized using the concordance index as defined above. For visualization and group-level testing, we generated Kaplan–Meier curves and log-rank tests for risk groups defined by a

single cutpoint selected by cross-validation in the TCGA and CPTAC training cohort, applied unchanged to the pooled validation samples after standardizing each linear predictor by the training mean and standard deviation. When multiple datasets were pooled, Cox models were stratified by dataset to allow cohort-specific baseline hazards. P-values for Kaplan–Meier group comparisons correspond to two-sided log-rank tests; continuous-score associations were evaluated using Cox models stratified by dataset.

## 8 Gene Program Correlation Analysis

To assess the biological identity of learned factors, we evaluated the correspondence between factor-specific gene rankings and established PDAC gene programs. Each column  $j$  of the gene program matrix  $W$  was first scaled by its column maximum,  $W_{ij}^* = W_{ij} / \max_i W_{ij}$ , placing all factors on a common  $[0, 1]$  scale, and the specificity score for gene  $i$  in factor  $j$  was defined as

$$s_{ij} = W_{ij}^* - \max_{j' \neq j} W_{ij'}^*.$$

Genes were ranked in descending order of  $s_{ij}$ . This ranking defines both the top- $n_{\text{top}}$  genes used to score samples (Supplementary Data: factor-specific gene lists) and, for the top 50 genes per factor, the loadings used to quantify factor–program correspondence.

For each factor  $j$  and each reference gene program  $G_k$ , we computed the Spearman rank correlation between the continuous factor loadings  $W_{\cdot j}$  (restricted to the union of the per-factor top-50 gene sets intersected with the reference programs) and a binary indicator vector  $\mathbf{1}[\text{gene} \in G_k]$  (main-text Fig. 2a,b; Methods).  $P$  values were derived from the asymptotic Spearman test and adjusted for multiple comparisons using the Benjamini–Hochberg procedure within each factor column independently. Reference programs were included in the visualization only if at least one factor yielded a Spearman correlation exceeding 0.2.

Reference gene programs displayed spanned classical and basal-like tumor signatures and activated/immune stromal programs from Moffitt et al.

<sup>15</sup>; classical and exocrine subtypes from Collisson et al.

<sup>20</sup>; classical-tumor, basal-like-tumor, exocrine, and activated-stroma compartment signatures from DECODER<sup>21</sup>; the pure classical, immune classical, desmoplastic, and basal-like tumor microenvironment programs from Puleo et al. <sup>17</sup>; the ECM and immune programs from Maurer et al. <sup>22</sup>; the iCAF cancer-associated fibroblast signature from Elyada et al. <sup>23</sup>; the proCAF, restCAF, and panCAF fibroblast subtypes from SCISSORS<sup>24</sup>; the proCAF and restCAF programs from the DeCAF study<sup>19</sup>; the classical A subtype from Chan-Seng-Yue et al.

<sup>25</sup>; and the ADEX (aberrantly differentiated endocrine exocrine) program from Bailey et al. <sup>16</sup>. Bailey’s ADEX program is retained because it is unique, whereas redundant Bailey programs, periductal subtypes, and MSI subtypes were dropped as exact duplicates of, or low-correlation relative to, other retained programs.

## 9 Reconstruction and Survival Contribution per Factor

NMF does not provide an orthogonal variance decomposition; it seeks a low-rank reconstruction of the expression matrix  $X \approx WH$ . We therefore measure each factor’s contribution to reconstruction rather than to “variance explained.” For a subset of factors  $S \subseteq \{1, \dots, k\}$ , define the reconstruction value

$$v(S) = \|X\|_F^2 - \left\| X - \sum_{j \in S} W_{\cdot j} H_{j \cdot} \right\|_F^2,$$

the reduction in residual error relative to the null (zero-factor) model, where  $W_{\cdot j} H_{j \cdot}$  is the rank-one outer-product contribution of factor  $j$  (the outer product of column  $j$  of  $W$  with row  $j$  of  $H$ ). The reconstruction Shapley value of factor  $j$ ,

$$\phi_j = \sum_{S \subseteq \{1, \dots, k\} \setminus \{j\}} \frac{|S|! (k - |S| - 1)!}{k!} (v(S \cup \{j\}) - v(S)),$$

is its average marginal reconstruction gain over all orderings of the factors. By the Shapley efficiency axiom,  $\sum_j \phi_j = v(\{1, \dots, k\})$ , the total reconstruction gain of the full model, so the normalized shares  $\phi_j / \sum_l \phi_l$  form an additive decomposition of reconstruction that sums to one; individual shares may be signed in principle but were all positive in the reported PDAC models. This is a genuine additive decomposition, in contrast to per-factor variance fractions, which for non-orthogonal factorizations such as NMF do not partition (cross-terms between rank-one contributions are non-zero). The Shapley shares are computed exactly by enumerating the  $2^k$  subsets (trivial at the small ranks considered here).

The survival contribution of factor  $j$  was quantified as the Type III partial likelihood increment: the reduction in Cox partial log-likelihood when factor  $j$  is omitted from the full  $k$ -factor model,

$$\Delta\ell_j = \ell_{\text{full}}(\hat{\beta}_{\text{full}}) - \ell_{-j}(\hat{\beta}_{-j}),$$

where  $\ell_{\text{full}}(\hat{\beta}_{\text{full}})$  is the maximized unpenalized Cox partial log-likelihood of the full  $k$ -factor model and  $\ell_{-j}(\hat{\beta}_{-j})$  is that of the reduced model refit on the remaining  $k - 1$  factors after omitting factor  $j$ , each evaluated at its own maximum partial likelihood estimate. Larger  $\Delta\ell_j$  indicates a greater marginal contribution of factor  $j$  to survival prediction given all other factors. Unlike the reconstruction Shapley shares, these Type III increments are conditional (each is evaluated against the full complement of the remaining  $k - 1$  factors) and are nonadditive: they do not partition the total model partial likelihood and are not comparable across factors as variance-style shares. We therefore report them as per-factor conditional increments rather than as an additive survival decomposition. These two quantities are plotted against each other in main text, Fig. 2c. In PDAC at  $k = 3$ , the reconstruction shares are 41.8%/18.6%/39.7% for NMF (N1/N2/N3) and 18.8%/25.2%/56.0% for DeSurv (D1/D2/D3), while the survival contribution  $\Delta\ell$  concentrates in DeSurv D1 ( $\Delta\ell = 66.4$ ) with all NMF factors negligible. The factors that contribute most to reconstruction are not the most prognostic.

## 10 Factor Correspondence Analysis

To assess which biological programs are preserved, altered, or suppressed under survival supervision, we quantified the pairwise correspondence between NMF and DeSurv factor loadings. For each pair of factors ( $j_{\text{NMF}}, j_{\text{DeSurv}}$ ), we computed the Spearman rank correlation between the corresponding columns of the NMF gene program matrix  $W_{\text{NMF}}$  and the DeSurv gene program matrix  $W_{\text{DeSurv}}$  over all genes. We use the full loading profile rather than a top-gene subset, which would impose an arbitrary threshold and overstate apparent preservation. Spearman correlation was used to capture monotone correspondence in gene loading ranks while remaining robust to scale differences between the two methods. Results are displayed as a heatmap with NMF factors as rows and DeSurv factors as columns (main text, Fig. 2d).

### 10.1 Cross-method reconstructability and reconstruction fidelity

The correlation heatmap summarizes pairwise similarity; to make a direction-aware statement of how much of each factor is *linearly reconstructable* from the other method’s basis, we regressed each factor’s gene-loading vector onto the full set of the other method’s loading vectors and report the ordinary least-squares  $R^2$  (Table S2). A high  $R^2$  means the factor is well reconstructed as a linear combination of the other method’s factors; a low  $R^2$  means it is not linearly reconstructable from that basis. This is a statement about linear reconstructability of gene-loading vectors, not about which method discovered a program first.

The headline is DeSurv D1:  $R^2 = 0.09$  from the NMF basis, indicating that its gene-loading vector is not well reconstructed as a linear combination of the matched-rank NMF factors. The biologically dominant tumor (N1) and stromal (N3) programs, by contrast, are largely reconstructable from the DeSurv basis ( $R^2 = 0.90$  and  $0.88$ ), so most of the unsupervised structure is retained; the principal reorganization is that the DeSurv basis resolves tumor signal into basal- and classical-aligned programs while reconstructing the exocrine factor (N2) less well ( $R^2 = 0.38$ ).

At the whole-matrix level, this reorganization carries a real but bounded reconstruction cost (Table S3). On the conventional centered  $R^2$ , DeSurv reconstructs 0.75 of the analysis matrix against 0.81 for unsupervised NMF, a difference of 6.0 percentage points.

The reorganization therefore has a real price. On the uncentered scale used by the reconstruction helper the gap is only 1.5 percentage points, but that scale flattens any nonnegative low-rank fit to a rank matrix, and on the conventional centered definition DeSurv gives up 6.0 percentage points of reconstruction  $R^2$  relative to unsupervised NMF (Table S3), a relative loss of about 7%. We report the centered figure in the main text.

## 10.2 The D1 axis is recovered by survival supervision, not by DeSurv specifically

If the D1 tumor–stroma axis were an artifact of DeSurv’s particular optimizer, independent survival-supervised methods should not recover it. We therefore fit three other supervised dimension-reduction methods on the same training cohort ( $n = 273$ , 139 events), namely supervised principal components (Cox-screened genes followed by PCA), one-component Cox partial least squares against martingale residuals, and a sparse Cox model (lasso-penalized, 32 selected genes), and compared their leading survival-aligned direction to the DeSurv D1 score. All three recovered D1 closely (per-patient  $|r| = 0.81$  for supervised PCA, 0.83 for Cox-PLS, and 0.85 for sparse Cox; mean 0.83), and each aligned specifically with D1 rather than the stromal D2 or basal-like D3 programs (Fig. S2). By contrast, the leading *unsupervised* direction (the first principal component, a variance-dominant unsupervised direction) was nearly orthogonal to D1 ( $|r| = 0.10$ ) and instead tracked the high-variance D2/D3 stroma–basal axis. This D1 axis is thus a property of survival supervision shared across methods, consistent with its being unreconstructable from the unsupervised NMF basis (Table S2) and with standard NMF’s failure to enrich for the proCAF stroma at matched rank, and not an idiosyncrasy of the DeSurv algorithm.

## 10.3 Robustness to tuned supervised comparators and compartment attribution

The recovery above establishes the D1 axis as a genuine survival direction shared across methods; here we ask what DeSurv adds beyond recovering it, using canonical, independently tuned comparators in place of the simple single-component versions above: supervised principal components (**superpc**; Bair–Tibshirani, with cross-validated screening threshold and component number) and penalized (lasso) Cox (**glmnet**, a single  $\lambda_{\min}$  model). Cox partial least squares was not re-run in this pass, as a validated multi-component implementation was unavailable in our environment; the two canonical methods are sufficient. Every method began from the same 1,970-gene training-only expression/variance universe and performed its own feature selection; DeSurv was scored with its Bayesian-optimization-selected top-270-per-factor programs; all scoring used the frozen, rank-transformed transfer with no validation refitting.

Prediction parity was preserved: the pooled frozen-transfer C-index was 0.61 for the DeSurv linear predictor, 0.60 for supervised PCA, and 0.60 for penalized Cox. Consistent with the score-level recovery above, both tuned methods aligned with the D1 direction (training  $|r| = 0.71$  for penalized Cox and 0.46 for supervised PCA) and with the DeSurv linear predictor ( $|r| = 0.86$  for penalized Cox and 0.65 for supervised PCA), but were essentially orthogonal to the activated-stroma/proCAF program D2 ( $|r| = 0.06$  for penalized Cox and 0.14 for supervised PCA): a single supervised risk axis captures the dominant tumour-axis signal but not the separate stromal program.

The compartment attribution made this explicit. Because D1 runs from a classical-tumour, restraining-stroma protective pole to a basal-like high-risk pole, a sparse supervised risk score aligned with D1 is naturally composed of basal-like tumour markers. Accordingly, across 12 cross-validation seeds the genes selected by both methods were reproducibly enriched (hypergeometric  $P < 0.01$ ) for the basal-like tumour program (in 12/12 seeds for penalized Cox and 12/12 for supervised PCA) and in no seed for the activated-stroma/proCAF, restraining-stroma, immune, or exocrine compartments. The supervised risk scores did nonetheless retain some stromal signal: each correlated modestly with an external activated-stroma/proCAF signature ( $|r| = 0.20$  for penalized Cox and 0.23 for supervised PCA), but well below DeSurv’s D2 program ( $|r| = 0.55$ ), which tracks that signature directly. The supervised risk scores retained modest stromal signal but did not resolve it as a

distinct program. DeSurv instead placed the dominant survival-associated direction within a three-program representation containing separately measurable classical/restCAF-like, proCAF-like and basal-like programs. Thus, the distinction is representational: DeSurv preserves stromal context as an explicit program even when that program does not contribute independently to the fitted linear predictor.

## Proper-scoring sensitivity of the external validation

The main-text external validation reports discrimination with Harrell’s C-index under a Cox proportional-hazards model. Because Harrell’s C can be optimistic when hazards are non-proportional<sup>26,27</sup>, we additionally evaluated the DeSurv linear predictor, its direction frozen from training, with a proper score, the integrated Brier score (IBS), in each validation cohort, recalibrating only its scalar slope and baseline hazard within each cohort and scoring over that cohort’s follow-up. For a single time-invariant risk score under proportional hazards the monotone score-to-survival map preserves the concordance ordering, so Antolini’s time-dependent concordance coincides with Harrell’s C; the additive quantity is therefore the proper score, not a second concordance. Across the 5 cohorts the DeSurv predictor achieved an event-weighted integrated Brier score of 0.159 versus 0.168 for a Kaplan–Meier reference, an index of prediction accuracy ( $IPA = 1 - IBS/IBS_{\text{null}}$ ) of 5.4% (per-cohort range 2.9–9.4%). Because each cohort’s IBS is integrated over its own event-time horizon, this pooled value is a descriptive event-weighted summary rather than a single common-horizon proper score, and the per-cohort IPA range is the more directly interpretable quantity. Two caveats bound how this analysis should be read. First, the recalibration step estimates both the scalar slope and the baseline hazard from each cohort’s own outcomes, so unlike the C-index and hazard-ratio analyses, which use a fully frozen predictor, this sensitivity evaluates the transported score direction under local recalibration rather than a frozen absolute-risk model. Second, the scores are computed on the same data used to fit that recalibration, with no cross-validation or resampling, so the reported IBS and IPA are apparent in-sample values and are optimistic. Both points cut against the reported improvement rather than for it, and the improvement is already small. This modest proper-score improvement over the null is consistent with the moderate discrimination ( $C = 0.61$ ) and with DeSurv’s contribution being a separately resolved tumour- and stroma-associated decomposition rather than higher predictive accuracy.

## Proportional-hazards diagnostic

Because the pooled hazard ratio assumes proportional hazards, we tested the scaled Schoenfeld residuals of the stratified validation model ( $\text{Surv}(t, \delta) \sim \text{risk\_z} + \text{strata}(\text{dataset})$ ) with `survival::cox.zph`. The proportional-hazards assumption was not supported for the standardized DeSurv linear predictor ( $\chi^2 = 14.5$ ,  $df = 1$ ,  $P = 1.4e - 04$ ), with the score’s effect attenuating over follow-up rather than reversing. The reported pooled hazard ratio should therefore be read as a time-averaged summary of a consistently prognostic score rather than a constant multiplicative effect; this time-varying behavior is precisely why we additionally report the horizon-integrated proper-scoring (IBS/IPA) sensitivity above, which integrates prediction accuracy over each cohort’s event-time horizon (to the 80th percentile of event times) rather than through a single hazard-ratio summary.

## Independence of discovery and validation

To make the separation between discovery, adjustment, and validation unambiguous: DeSurv is trained with survival supervision only; it never uses PurIST, DeCAF, or any molecular subtype label during factorization or hyperparameter selection. The PurIST and DeCAF subtype calls were fixed before the DeSurv analysis (loaded as precomputed per-cohort annotations) and enter only as covariates in the adjusted Cox models, not as inputs to DeSurv. All rank and hyperparameter selection (including the one-standard-error rule) was performed by cross-validation entirely within the TCGA and CPTAC training data; no external validation cohort was used for training, cutpoint selection, or tuning. Validation-cohort expression enters only at scoring, a fixed-basis projection ( $Z = W^\top X_{\text{new}}$ ) that uses no validation outcome, and validation-cohort survival outcomes were used only for the final, out-of-sample performance evaluation.

Adjustment for clinical covariates and tumor purity

## Part II

# Supplementary Results

## 11 Empirical optimization stability across random initializations

The idealized alternating scheme is not accompanied by a formal convergence guarantee for the fitted models, both because the reported analyses set the loadings ridge penalty to zero ( $\lambda_H = 0$ ) and because the implemented algorithm combines column normalization, gradient balancing between the reconstruction and Cox terms, and a Coxnet approximation of the  $\beta$  update. We therefore evaluated optimization stability empirically across  $R = 100$  independent random initializations of the selected PDAC model. The stopping tolerance was  $10^{-7}$  with a maximum of 4,000 iterations. Of the 100 runs, 100 terminated after satisfying the stopping criterion and 0 reached the iteration limit; the median number of iterations was 38 (range, 17–233). No numerical failures were observed. In every run the recorded full penalized objective decreased monotonically to a numerically stable value. Supplementary Fig. S3 shows five runs selected at random (seed 147). These analyses document empirical optimization stability for the reported application rather than providing a formal convergence guarantee.

## 12 Stability of the recovered gene programs

The previous section documents that the optimizer converges. It does not establish that the programs it converges to are the same ones each time, and that is the property the manuscript relies on when it applies a frozen decomposition across cohorts. Because we raise the non-identifiability of reconstruction-based factorization as a limitation of unsupervised approaches, the same standard should be applied here. We therefore measured stability at the level of the recovered programs rather than the objective, using the same  $R = 100$  cached initializations (`code/20`, `code/21`). No model was refitted for the restart analysis.

**Individual restarts are highly dispersed.** Comparing pairs of independent random restarts, and matching their columns one to one by absolute Spearman correlation of gene loadings, the median overlap between the top 270 genes of matched programs was 0.13 by Jaccard index, against a chance baseline of 0.074 for two independent draws of that size from 1,970 genes. Matched gene-loading correlations were moderate (0.59–0.60 across the three programs). Over the same runs, training concordance was nearly constant (median 0.704, standard deviation 0.010). Near-identical predictive performance from bases that share little gene membership is the signature of a non-identifiable objective, and it is precisely why the reported model is not a single run.

**The reported procedure is reproducible.** The reported basis is obtained by aggregating restarts into a consensus initialization and then fitting from it. To test whether that procedure returns the same programs, we partitioned the cached restarts into two disjoint halves, built a consensus initialization from each half independently, fitted from each at the production settings, and compared the results (8 random splits). Two independently seeded consensus fits agreed closely with each other (gene-loading correlation 0.93–0.95, patient-score correlation 0.97–0.99), and each agreed still more closely with the reported basis (gene-loading correlation 0.97–0.99, top-270 Jaccard 0.73–0.88). Splitting the restarts halves the number aggregated relative to the reported procedure, so these figures are conservative.

The two results should be read together. The optimizer alone is not identifiable, and the consensus step is what makes the reported decomposition a stable object rather than one draw from a wide distribution. Reproducibility claims in this manuscript refer to that end-to-end procedure.

**Stability under data perturbation is a separate and weaker property.** The analyses above vary the initialization while holding the data fixed. Refitting on repeated 80% subsamples of the training cohort and comparing top-100 gene sets instead gives a mean pairwise Jaccard of 0.34 for DeSurv, against 0.49 for Cox partial least squares, 0.38 for supervised principal components and 0.22 for sparse Cox. DeSurv is

mid-range. This comparison favours dense methods by construction, because a score that spreads weight across many genes has a more stable top-100 list than one built from a selective basis, as the sparse Cox value illustrates. We report it because gene-level resampling stability is a reasonable thing for a reader to ask about, not because it is the property the frozen-program argument rests on.

## 13 Simulation results under null and mixed scenarios

To characterize the operating characteristics of DeSurv across different signal regimes, we evaluated performance under three simulation scenarios. In the prognostic scenario (main text Fig. 4), prognostic programs explained low variance relative to outcome-neutral background signals, and survival depended on 150 factor-specific marker genes ( $\beta_1 = 2$ ,  $\beta_2 = \beta_3 = 0$ ). In the null scenario, survival times were generated independently of gene expression ( $\beta = 0$ ), so no prognostic structure existed. In the mixed scenario, survival depended on 300 genes per lethal factor, of which 50% were factor-specific markers and 50% were background genes with loadings shared across all factors. All scenarios used  $G = 3,000$  genes,  $N = 200$  samples,  $K = 3$  true factors, and identical gamma distribution parameters for the gene loading matrix  $W$  (see Materials and Methods). Performance was evaluated using cross-validated concordance index (C-index), precision of recovered prognostic gene sets, and the distribution of selected ranks across simulation replicates.

Under the null scenario (Fig. S4a–b), DeSurv showed no advantage over unsupervised NMF ( $\alpha = 0$ ): both methods yielded C-index values near 0.5, and selected ranks concentrated at low values (NMF favoring  $k = 2$ ), consistent with the absence of survival signal. DeSurv therefore did not recover spurious prognostic structure when none existed. Because the survival objective is uninformative under the null, the cross-validated concordance is essentially flat in  $\alpha$  and the Bayesian-optimized supervision strength is unidentified, dispersing across its admissible range rather than concentrating near zero (median  $\alpha = 0.59$ , IQR 0.19–0.82; range 0–0.94 across 100 replicates). The null-scenario specificity thus reflects the flat concordance surface itself, not the optimizer driving  $\alpha$  to zero.

Under the mixed scenario (Fig. S4c–f), DeSurv’s advantage was concentrated in marker-gene recovery. Precision against the factor-specific marker genes alone (150 genes; Fig. S4d left) substantially favored DeSurv, whereas precision against the full 300-gene survival set including background genes (Fig. S4d right) was more modestly improved, indicating similar background-gene recovery between methods. The learned factor aligned with the true marker program also received a larger Cox coefficient ( $|\beta|$ ) under DeSurv (Fig. S4e), and C-index was modestly higher (Fig. S4c). Across the three scenarios the pattern was graded: DeSurv’s advantage was greatest when variance and prognosis diverged (prognostic), attenuated when they partially overlapped (mixed, as background-gene dilution reduced the discrimination gain), and absent when no survival signal existed (null). DeSurv is therefore most useful when the dominant transcriptional axes are expected to be prognostically neutral, as in cancers with high tumor-purity variation or dominant tissue-composition signals.

## 14 Cross-validated C-index across factorization rank

To evaluate how the number of latent factors  $k$  affects prognostic performance, as a sensitivity analysis we performed an exhaustive grid search over  $k \in \{2, 3, \dots, 12\}$  and supervision strength  $\alpha \in \{0, 0.05, \dots, 0.95\}$  using 5-fold cross-validation on the TCGA+CPTAC training cohort. For each  $k$ , the best-performing  $\alpha$  was selected by maximizing the mean CV C-index. The unsupervised NMF baseline ( $\alpha = 0$ ) was evaluated at the same  $k$  values for comparison. Results are shown for two gene-selection settings:  $n_{\text{top}} = 270$  top genes per factor and  $n_{\text{top}} = \text{ALL}$  (all genes retained), both at  $\lambda = 0.349$  and  $\xi = 0.056$ . External validation C-index is shown descriptively; validation outcomes were not used to select  $k$ ,  $\alpha$ ,  $\lambda$ ,  $\xi$ , or  $n_{\text{top}}$ .

Fig. S5 shows the resulting C-index curves for both training (CV) and external validation (pooled across held-out cohorts), with shaded ribbons indicating  $\pm 1$  standard error. Rank was selected on the training cross-validated concordance: for DeSurv the curve is close to flat across the whole range examined, spanning

about two standard errors from  $k = 2$  to  $k = 12$  with its nominal maximum at  $k = 9$  rather than at the selected rank, and the one-standard-error rule accordingly selected the parsimonious  $k = 3$ . The selection depends on the gene-selection setting: under panel (b), with all genes retained, the same rule selects  $k = 2$ . DeSurv exceeded unsupervised NMF at matched rank at every  $k$  examined. The external-validation curve is shown only descriptively (validation outcomes were used for no part of rank or hyperparameter selection) and is consistent with this training-based choice, with DeSurv at  $k = 3$  attaining concordance comparable to standard NMF at higher ranks (e.g.,  $k = 7$ ; Table S5). We therefore read this as a training/cross-validation sensitivity analysis of rank, not as an external model-selection argument.

## 15 Standard NMF rank selection heuristics

Standard unsupervised rank selection criteria (reconstruction residuals, cophenetic correlation coefficient, and mean silhouette width) yielded inconsistent guidance for selecting the factorization rank  $k$  in the PDAC training cohort (TCGA + CPTAC). Reconstruction residuals decreased smoothly without a clear elbow (Fig. S6a). Cophenetic correlation fluctuated without a distinct transition point (Fig. S6b). Mean silhouette width favored very low ranks that conflicted with other criteria (Fig. S6c). These contradictions motivated the survival-supervised rank selection approach described in the main text.

## 16 Cutpoint selection and validation of dichotomized risk groups

To convert the continuous DeSurv linear predictor into a clinically interpretable binary risk stratification, we selected an optimal z-score cutpoint via cross-validation on the TCGA+CPTAC training cohort. For each candidate cutpoint in  $\{-2.0, -1.8, \dots, 2.0\}$ , we computed the mean absolute log-rank z-statistic across held-out folds and selected the cutpoint maximizing this metric (Fig. S7a). We then applied this cutpoint to external validation cohorts by standardizing each patient’s linear predictor using the training mean and standard deviation. Fig. S7b shows the resulting Kaplan–Meier curves for the pooled external validation samples. To respect between-cohort differences in baseline hazard, the reported inference is cohort-stratified: the hazard ratio is from a Cox model with the risk group as covariate and dataset as a stratifying factor ( $\text{Surv}(t, \delta) \sim \text{group} + \text{strata}(\text{dataset})$ ), and the accompanying log-rank test is stratified by dataset. The Kaplan–Meier curves themselves are shown pooled for visualization. Panels c–f display per-cohort validation KM curves for Dijk, Moffitt, PACA-AU (de-duplicated across array and RNA-seq platforms, retaining RNA-seq for dual-platform patients), and Puleo, respectively. Across cohorts, the DeSurv-derived high-risk group showed directionally shorter survival, indicating that the training-derived cutpoint generalizes to independent datasets.

## 17 Standard NMF factor structure at independently selected rank

Throughout the main text, both DeSurv and standard NMF were fit at the same rank ( $k = 3$ ), enabling a controlled comparison of the effect of survival supervision on factor structure. When standard NMF selects its own rank independently, Bayesian optimization of the unsupervised ( $\alpha = 0$ ) baseline selects  $k = 7$ , whereas supervised DeSurv BO selects  $k = 3$ . We examine here the NMF factor structure at this independently selected  $k = 7$  to illustrate the additional fragmentation that arises when supervision is absent (Fig. S8).

At BO-selected  $k = 7$  under  $\alpha = 0$  (Fig. S8), standard NMF shows increased fragmentation relative to the matched-rank  $k = 3$  solution: multiple factors partially overlap the same PDAC gene signatures, and no additional factor maps cleanly onto a distinct established PDAC program; the added factors fragment existing signatures rather than resolving new biology.

Standard NMF at elbow-selected  $k = 5$  and BO-selected  $k = 7$  both substantially improve over NMF at  $k = 3$  across all cohorts (Table S5); additional unsupervised factors capture prognostically relevant variation missed

at the matched rank. At  $k = 7$ , NMF matches or exceeds DeSurv’s per-cohort C-index in most cohorts, indicating that unsupervised factorization can approach supervised concordance given enough factors. This comes at the cost of a less parsimonious and less interpretable factorization: the  $k = 7$  NMF solution shows increased fragmentation (Fig. S8), distributing the same prognostic signal across a rank-sensitive set of overlapping factors rather than concentrating it. The DeSurv linear predictor, by contrast, retains significant prognostic value after adjustment for established classifiers with only three factors (main-text Table 1).

## Supplementary Data: factor-specific gene lists

As Supplementary Data, Tables S6–S8 list the top 270 genes for each of the three DeSurv factors, ranked by the factor specificity score  $s_{ij}$  defined in Supplementary Information, Section 8. The 270-gene cutoff corresponds to the Bayesian optimization–selected projection dimension ( $n_{\text{top}} = 270$ ); these are the genes used to compute sample-level factor scores for survival prediction. Factor D1 is enriched for classical tumor and restCAF-like stromal genes, Factor D2 for proCAF-associated stromal genes, and Factor D3 for basal-like tumor genes.

## 18 Translational transportability into independently treated PDAC cohorts

The programs were learned in treatment-naïve, non-metastatic PDAC, the setting in which the gene-program matrix  $W$  and Cox coefficients  $\beta$  were trained. As a translational test we projected the fixed programs, without retraining, into independently treated PDAC cohorts and asked whether they retained clinically relevant associations. We used two complementary resources. First, the O’Kane/COMPASS cohort of treated metastatic PDAC (laser-capture microdissected and therefore epithelial-enriched), which supports a transportability test of the tumor-associated  $D_1$  direction (Fig. 5a). Second, three bulk-tissue treated cohorts spanning two disease settings, the metastatic gemcitabine-backbone trials Rash ( $n = 85$ , gemcitabine + erlotinib) and Accept ( $n = 70$ , gemcitabine  $\pm$  afatinib) and the borderline-resectable / locally advanced Linehan cohort ( $n = 66$ ; FOLFIRINOX  $\pm$  the CCR2 inhibitor PF-04136309), in which, because tissue is bulk, both the PurIST (tumor-intrinsic) and DeCAF (stromal) classifiers are computable and the full three-program structure, including the stromal  $D_2$  program, can be evaluated (Fig. 5c; Table 1 gives the same dual adjustment used in the main validation). Because the cohorts differ in stage and regimen, they are analysed per cohort and never pooled. The elastic-net penalty shrank the trained survival coefficient for the proCAF program  $D_2$  to exactly zero in the treatment-naïve model (an estimated selection outcome, not an imposed constraint), so any association of  $D_2$  in the treated cohorts reflects a marginal re-evaluation of the projected program in a new clinical context rather than a property of the trained predictor.

### 18.1 Prognostic transportability of D1 in treated metastatic disease

To evaluate transportability in treated metastatic disease without combining treatment arms, we projected the frozen DeSurv programs into the O’Kane/COMPASS cohort and estimated  $D_1$  associations separately within each arm.  $D_1$  was associated with lower mortality in the same direction in FOLFIRINOX (HR per SD 0.80 (0.61–1.03)), gemcitabine/nab-paclitaxel (HR 0.57 (0.37–0.86)) and the GA/experimental arm (HR 0.83 (0.58–1.20); Fig. 5a). An arm-stratified Cox model allowing treatment-specific baseline hazards yielded a common association of HR 0.75 (0.62–0.90) ( $P = 0.003$ ; 173 patients, 123 deaths), with no evidence of heterogeneity across treatment arms (interaction  $P = 0.40$ ). Because these specimens were epithelial-enriched by laser-capture microdissection, this analysis specifically tests transportability of the tumor-associated  $D_1$  direction rather than the stromal programs.

## 18.2 Program associations in bulk treated cohorts

Factor scores were computed by projection using the same trained, top-gene-truncated  $\widetilde{W}$  ( $n_{\text{top}} = 270$  per factor, union across factors) and within-sample rank convention as the main-paper validation cohorts ( $Z_{\text{new}} = \widetilde{W}^T X_{\text{new}}$ ), z-standardized within cohort; results were near-identical under a full-gene projection (Spearman  $r \geq 0.86$ ), so the treated-cohort findings do not depend on the projection convention. Because the three treated cohorts differ in disease setting and regimen (Linehan borderline-resectable/locally-advanced on FOLFIRINOX $\pm$ CCR2 inhibitor; Rash and Accept metastatic on gemcitabine backbones), and  $D_1$ , the classical/restCAF-aligned program of the main text, indexes the basal–classical contrast in these cohorts (higher  $D_1$  marks classical tumors, lower  $D_1$  basal ones) and is heterogeneous across them, we report each cohort separately rather than as a pooled estimate.  $D_1$  is reported *marginally*: because  $D_1$  tracks the basal–classical contrast and is collinear with the PurIST classifier (it separates PurIST calls at AUC 0.84–0.89), adjusting  $D_1$  for PurIST conditions the same axis on itself and over-adjusts; the marginal estimate is the appropriate measure of its prognostic value.  $D_2$  (a stromal program, not strongly collinear with the classifiers) is reported both marginal and PurIST + DeCAF-adjusted.

The proCAF stromal program  $D_2$  was protective in all three cohorts (both marginally and after adjustment, with significance reached in Linehan), with a consistent direction across the independent settings and chemotherapy backbones; it was significant in Linehan and non-significant but directionally concordant in the metastatic cohorts (Accept and Rash). Because  $D_2$  is stromal and not strongly collinear with the classifiers, its association was not an artifact of adjustment: in Linehan it was significant both before and after adjustment, and across all three cohorts the adjusted estimate strengthened rather than attenuated relative to the marginal one, indicating that the  $D_2$  association was not explained by PurIST or DeCAF adjustment.  $D_1$  was prognostic in the less-advanced Linehan cohort (marginal HR 0.58 (0.36–0.94)) but not in the metastatic cohorts, consistent with attenuation of the basal–classical axis itself with advancing stage rather than a failure of  $D_1$ , since PurIST, Moffitt and Bailey show the same attenuation in these cohorts. This reading depends on the reporting convention set out above, and we state the dependence rather than leave it implicit. Under classifier adjustment the pattern reverses, with Linehan  $D_1$  attenuating to 0.83 (0.48–1.45), 0.517 and Rash  $D_1$  strengthening to 0.61 (0.43–0.88), 0.007, while Accept remains null in both models. We report  $D_1$  marginally because it is collinear with PurIST in these cohorts, so the adjusted estimate removes much of the axis it is meant to measure, but neither convention supports a claim stronger than setting-dependence.  $D_3$  (basal-like) was not associated with overall survival in any treated cohort, consistent with the basal–classical contrast residing in  $D_1$ .

## 18.3 Treatment-associated change in D2

In the 29 Linehan patients with paired pre- and post-treatment biopsies,  $D_2$  increased significantly from pre- to post-treatment (paired Wilcoxon 0.013; mean  $\Delta = +0.34$  SD; Fig. 5b), consistent with treatment-associated remodeling of the proCAF program.

## 18.4 Arm-separated exploratory analyses

We also examined each treatment arm separately across overall survival, time to progression, binary response and continuous depth of response (Supplementary Table S10). These within-arm comparisons are underpowered for between-arm (treatment-specific) contrasts: the arms are small, and the few nominal associations were scattered across programs and endpoints and did not survive correction for multiple testing. The prognostic association instead prevails in the better-powered pooled and arm-stratified analyses (the O’Kane arm-stratified D1 effect and the consistent protective direction of D2 across the bulk cohorts), whereas differential effects by treatment arm are not resolvable in these data.

{Disease-control (CR/PR/SD vs PD) analyses were identical to the binary response analyses because no complete responses occurred in these treatment arms, and are therefore not reported separately or counted twice in the multiplicity adjustment. These analyses were exploratory: nominal  $P$  values are reported for transparency, but no association remained significant after correction for multiple testing, and estimates from the smallest treatment arms are imprecise and should not be interpreted as evidence of treatment-specific effects.}

## 18.5 Caveats

These analyses are exploratory and prognostic, not predictive: treatment, stage and cohort are confounded, the treated cohorts are modestly sized, and no analysis plan was preregistered. The programs are analysed per cohort and never pooled, and  $D_1$  is reported marginally in the bulk cohorts because it is collinear with the PurIST classifier. Robust treated-cohort validation, and any assessment of whether the programs predict differential treatment benefit, will require larger harmonized series with adequately powered treatment-by-program interaction analyses.

## SI References

1. Lee, D. D. & Seung, H. S. Algorithms for non-negative matrix factorization. in *Advances in neural information processing systems 13 (NIPS 2000)* (eds. Leen, T. K., Dietterich, T. G. & Tresp, V.) 556–562 (MIT Press, 2001).
2. Pascual-Montano, A., Carazo, J. M., Kochi, K., Lehmann, D. & Pascual-Marqui, R. D. Nonsmooth nonnegative matrix factorization (nsNMF). *IEEE Transactions on Pattern Analysis and Machine Intelligence* **28**, 403–415 (2006).
3. Simon, N., Friedman, J. H., Hastie, T. & Tibshirani, R. Regularization paths for cox’s proportional hazards model via coordinate descent. *Journal of Statistical Software* **39**, 1–13 (2011).
4. Donoho, D. & Stodden, V. When does non-negative matrix factorization give a correct decomposition into parts? in *Advances in neural information processing systems 16 (NIPS 2003)* (eds. Thrun, S., Saul, L. K. & Schölkopf, B.) 1141–1148 (MIT Press, 2004).
5. Laurberg, H., Christensen, M. G., Plumbley, M. D., Hansen, L. K. & Jensen, S. H. Theorems on positive data: On the uniqueness of NMF. *Computational Intelligence and Neuroscience* **2008**, 764206 (2008).
6. Gillis, N. The why and how of nonnegative matrix factorization. in *Regularization, optimization, kernels, and support vector machines* (eds. Suykens, J. A. K., Signoretto, M. & Argyriou, A.) 257–291 (Chapman; Hall/CRC, 2014).
7. Brunet, J.-P., Tamayo, P., Golub, T. R. & Mesirov, J. P. Metagenes and molecular pattern discovery using matrix factorization. *Proceedings of the National Academy of Sciences USA* **101**, 4164–4169 (2004).
8. Kim, J. & Park, H. *Sparse Nonnegative Matrix Factorization for Clustering*. <https://faculty.cc.gatech.edu/~hpark/papers/GT-CSE-08-01.pdf> (2008).
9. Gillis, N. & Glineur, F. Accelerated multiplicative updates and hierarchical ALS algorithms for nonnegative matrix factorization. *Neural Computation* **24**, 1085–1105 (2012).
10. Cai, D., He, X., Han, J. & Huang, T. S. Graph regularized nonnegative matrix factorization for data representation. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **33**, 1548–1560 (2011).
11. Blei, D. M. & McAuliffe, J. D. Supervised topic models. in *Advances in neural information processing systems 20 (NIPS 2007)* (eds. Platt, J. C., Koller, D., Singer, Y. & Roweis, S. T.) 121–128 (Curran Associates, Inc., 2007).
12. Raphael, B. J. *et al.* Integrated genomic characterization of pancreatic ductal adenocarcinoma. *Cancer Cell* **32**, 185–203 (2017).
13. Cao, L. *et al.* Proteogenomic characterization of pancreatic ductal adenocarcinoma. *Cell* **184**, 5031–5052 (2021).
14. Dijk, F. *et al.* Unsupervised class discovery in pancreatic ductal adenocarcinoma reveals cell-intrinsic mesenchymal features and high concordance between existing classification systems. *Scientific Reports* **10**, 337 (2020).
15. Moffitt, R. A. *et al.* Virtual microdissection identifies distinct tumor- and stroma-specific subtypes of pancreatic ductal adenocarcinoma. *Nature Genetics* **47**, 1168–1178 (2015).
16. Bailey, P. *et al.* Genomic analyses identify molecular subtypes of pancreatic cancer. *Nature* **531**, 47–52 (2016).
17. Puleo, F. *et al.* Stratification of pancreatic ductal adenocarcinomas based on tumor and microenvironment features. *Gastroenterology* **155**, 1999–2013 (2018).
18. Rashid, N. U. *et al.* Purity independent subtyping of tumors (PurIST), a clinically robust, single-sample classifier for tumor subtyping in pancreatic cancer. *Clinical Cancer Research* **26**, 82–92 (2020).
19. Peng, X. L. *et al.* DeCAF defines clinical fibroblast subtypes and multidimensional tumor-stroma crosstalk shaping prognosis and immunotherapy response. *Cell Reports Medicine* **7**, 102611 (2026).
20. Collisson, E. A. *et al.* Subtypes of pancreatic ductal adenocarcinoma and their differing responses to therapy. *Nature Medicine* **17**, 500–503 (2011).
21. Peng, X. L., Moffitt, R. A., Torphy, R. J., Volmar, K. E. & Yeh, J. J. De novo compartment deconvolution and weight estimation of tumor samples using DECODER. *Nature Communications* **10**, 4729 (2019).
22. Maurer, C. *et al.* Experimental microdissection enables functional harmonisation of pancreatic cancer subtypes. *Gut* **68**, 1034–1043 (2019).
23. Elyada, E. *et al.* Cross-species single-cell analysis of pancreatic ductal adenocarcinoma reveals antigen-presenting cancer-associated fibroblasts. *Cancer Discovery* **9**, 1102–1123 (2019).
24. Leary, J. R. *et al.* Sub-cluster identification through semi-supervised optimization of rare-cell silhouettes (SCISSORS) in single-cell RNA-sequencing. *Bioinformatics* **39**, btad449 (2023).

- 25. Chan-Seng-Yue, M. *et al.* Transcription phenotypes of pancreatic cancer are driven by genomic events during tumor evolution. *Nature Genetics* **52**, 231–240 (2020).
- 26. Birolo, G. *et al.* Beyond Cox models: Assessing the performance of machine-learning methods in non-proportional hazards and non-linear survival analysis. *Computers in Biology and Medicine* **198**, 111176 (2025).
- 27. Hao, L., Kim, J., Kwon, S. & Ha, I. D. Deep learning-based survival analysis for high-dimensional survival data. *Mathematics* **9**, 1244 (2021).

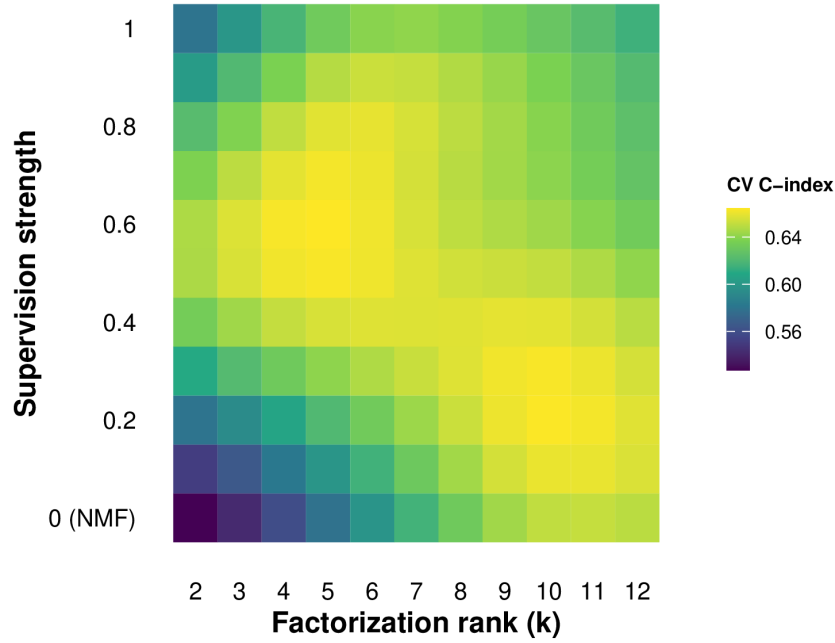


Figure S1: Bayesian-optimization tuning surface for DeSurv in the PDAC training cohort (TCGA + CPTAC). Gaussian-process-predicted mean cross-validated C-index over factorization rank ( $k$ ) and supervision strength ( $\alpha$ ;  $\alpha = 0$  corresponds to standard NMF). The response surface shows a broad region of high cross-validated performance at intermediate supervision and moderate rank, with discrimination degrading toward the unsupervised ( $\alpha = 0$ ) boundary. This is empirical model-selection tuning on the training data, not a simulation result.

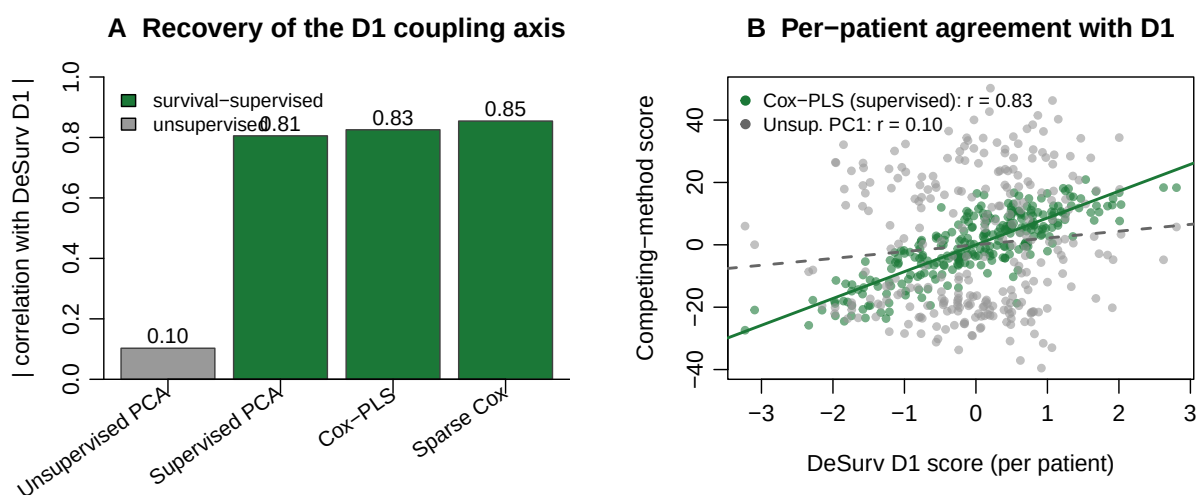


Figure S2: Independent survival-supervised methods recover the D1 axis; an unsupervised method does not. (A) Absolute correlation (across training-cohort patients) of each method's leading direction with the DeSurv D1 score, in the training cohort: supervised principal components, Cox partial least squares, and sparse Cox all recover D1 (green), whereas the leading unsupervised principal component does not (grey). (B) Sample-level agreement with D1 for Cox-PLS (supervised, green) versus the first unsupervised principal component (grey).

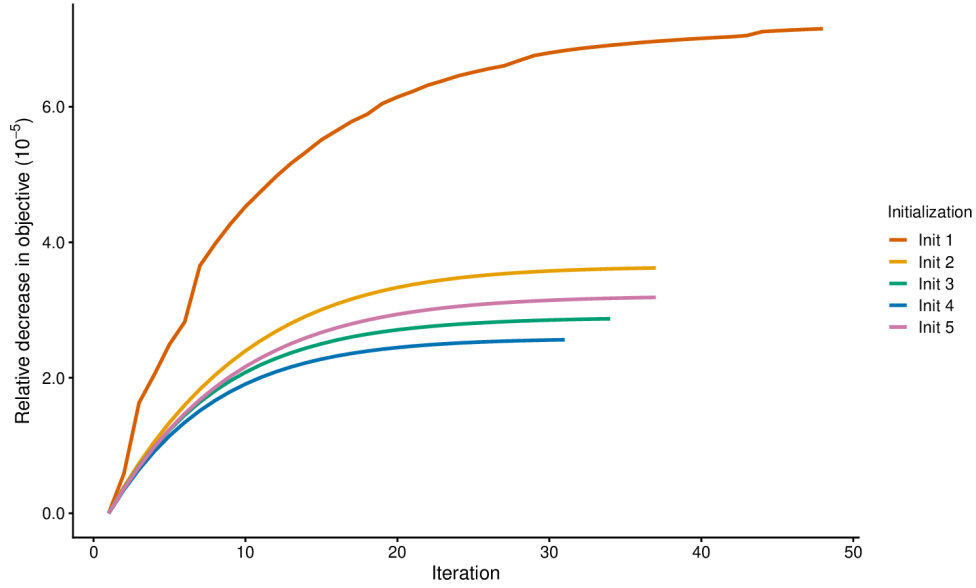


Figure S3: Empirical optimization-stability diagnostic for the DeSurv objective across five randomly selected multi-start initializations on the PDAC training cohort (TCGA + CPTAC). Each line shows the relative decrease in the objective from its initial value,  $(\ell_0 - \ell_t)/\ell_0$  (in units of  $10^{-5}$ ; the reconstruction-dominated total objective changes by a small relative amount), as a function of iteration number. In these runs the recorded full objective decreases monotonically and stabilizes within the iteration budget. This figure is an empirical diagnostic of optimization behavior for the reported application rather than a formal convergence proof.

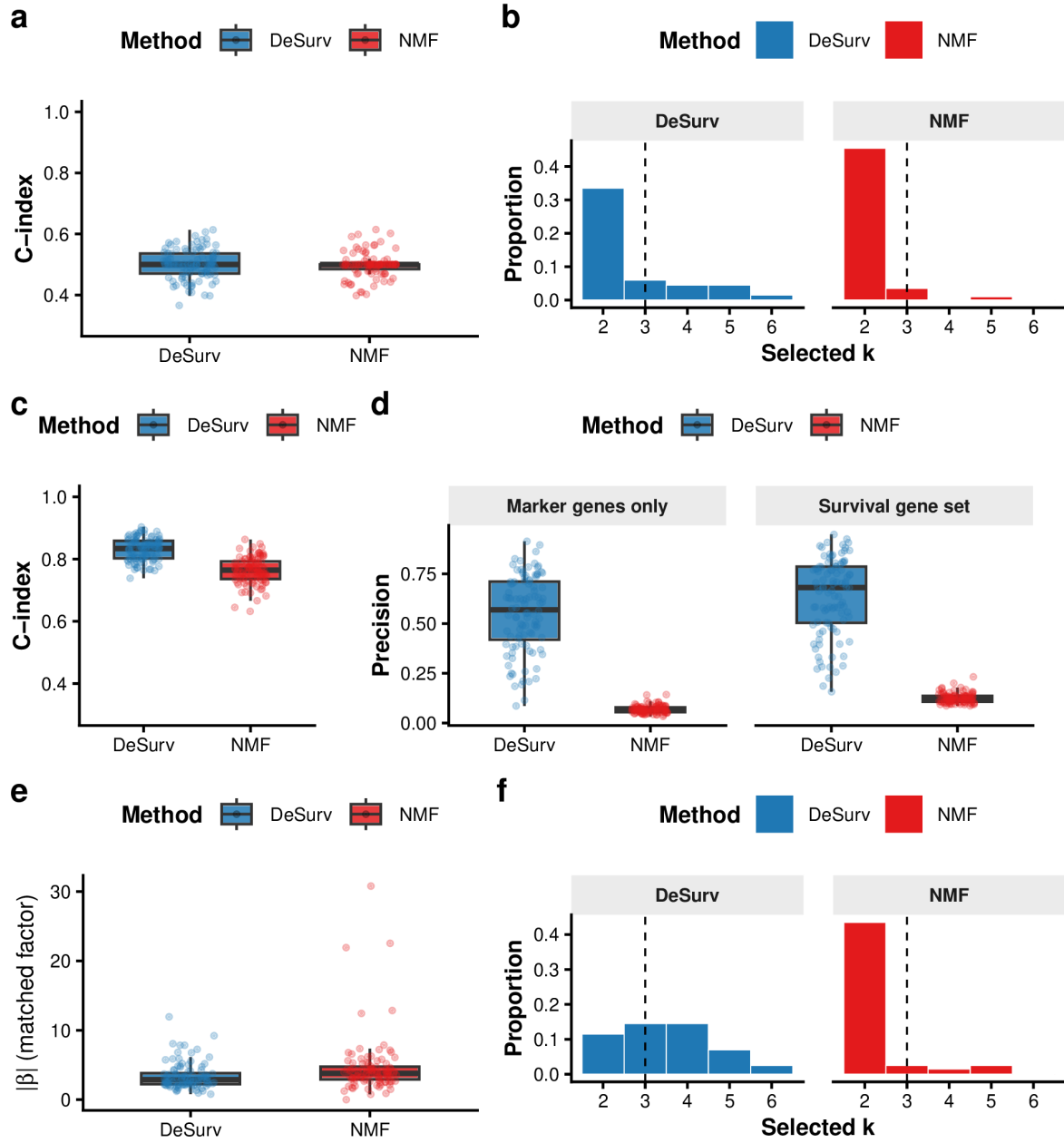


Figure S4: DeSurv performance under null and mixed simulation scenarios. (a–b) Null scenario ( $\beta = 0$ ): both methods yield C-index  $\approx 0.5$  (a) and comparable rank distributions (b), confirming DeSurv does not recover spurious prognostic structure; precision is omitted as no true survival genes exist. (c–f) Mixed scenario: survival depends on factor-specific marker genes (50%) and shared background genes (50%). (c) C-index is modestly higher for DeSurv. (d) Precision by gene-set definition: marker-only precision (150 genes; left facet) substantially favors DeSurv; full 300-gene survival-set precision (right facet) shows a more modest improvement. (e) The matched factor receives a substantially larger Cox coefficient  $|\beta|$  under DeSurv, confirming preferential amplification of marker-specific signal. (f) Selected-rank distributions. DeSurv (blue); NMF (red,  $\alpha = 0$ ); both Bayesian-optimized.

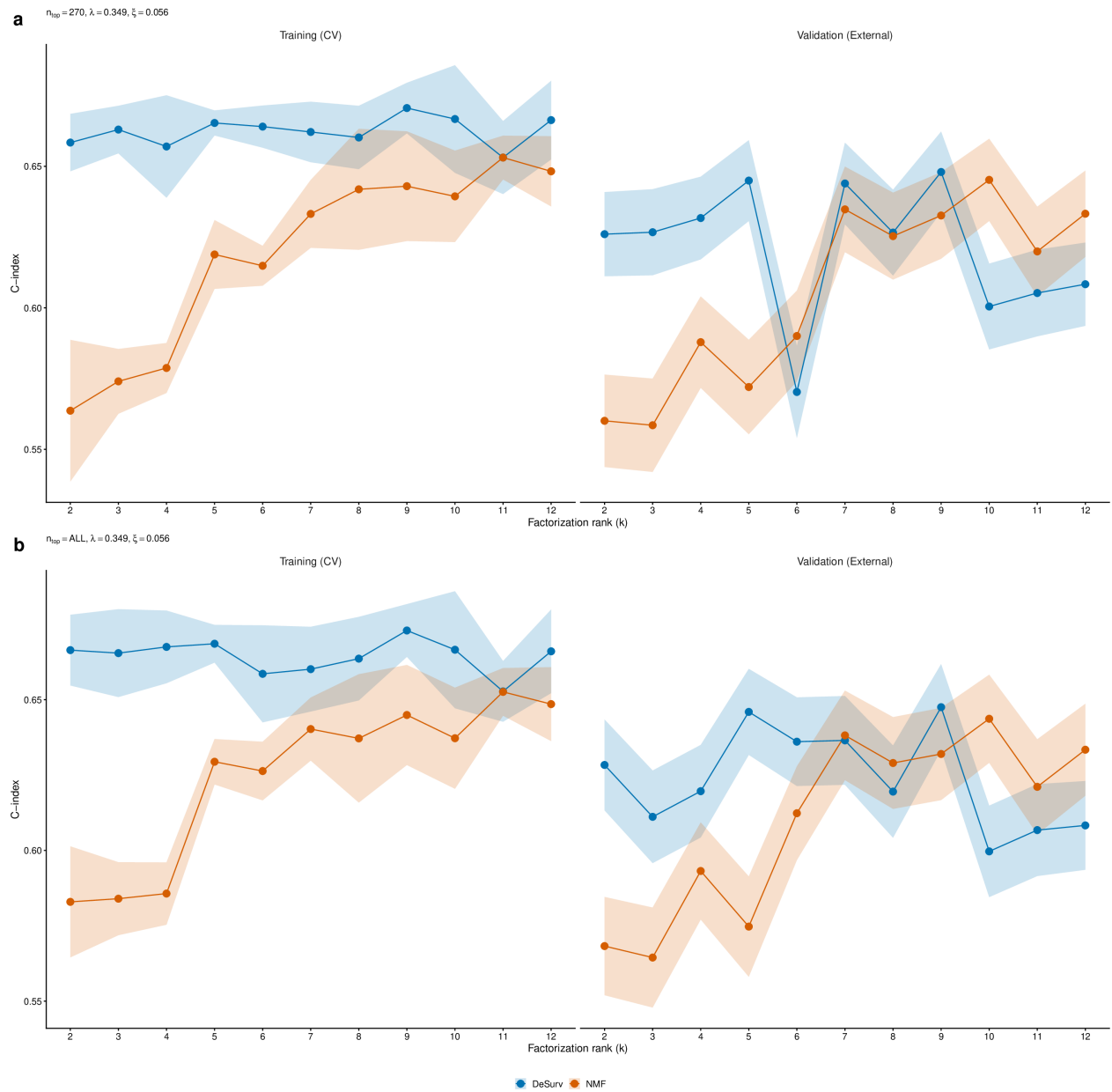


Figure S5: Training (CV) and external validation C-index as a function of factorization rank  $k$  for DeSurv (blue) and standard NMF (red,  $\alpha = 0$ ). Shaded ribbons show  $\pm 1$  SE. For each  $k$ , the best  $\alpha$  was selected by maximizing the mean CV C-index. Validation C-index is pooled across all external cohorts. (a) Results using  $n_{\text{top}} = 270$  top genes per factor ( $\lambda = 0.349, \xi = 0.056$ ). (b) Results retaining all genes ( $n_{\text{top}} = \text{ALL}; \lambda = 0.349, \xi = 0.056$ ).

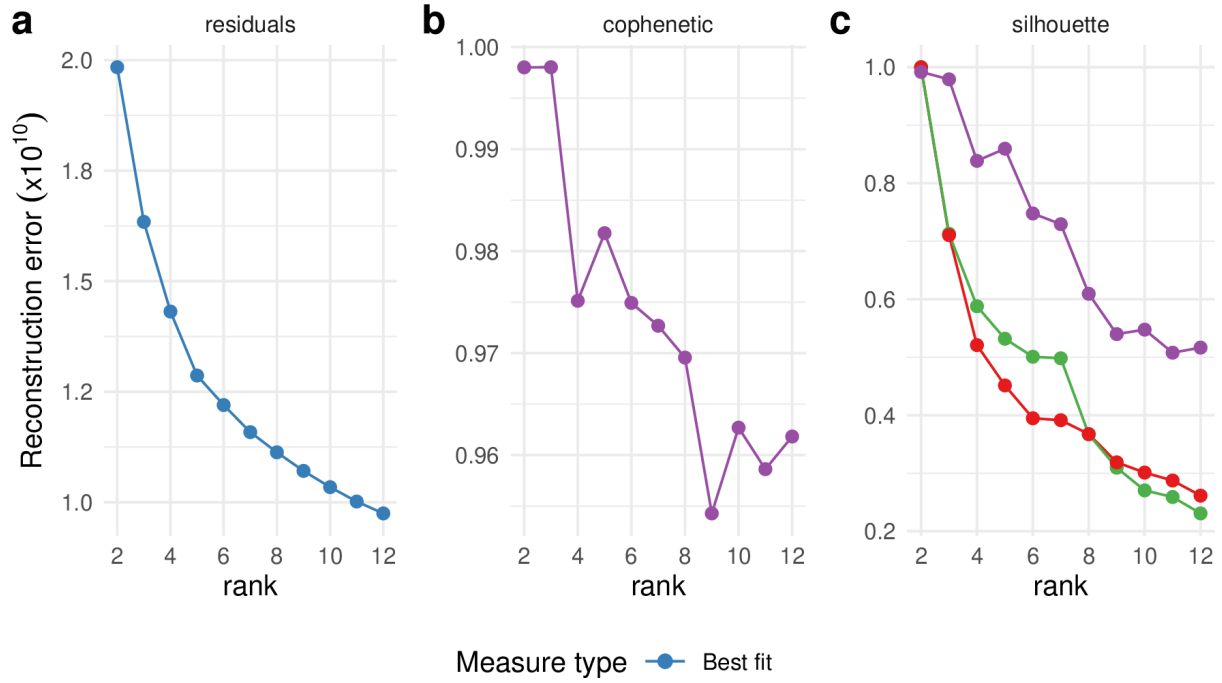


Figure S6: Standard unsupervised NMF rank selection heuristics for the PDAC training cohort (TCGA + CPTAC). (a) Reconstruction residuals as a function of factorization rank  $k$ . (b) Cophenetic correlation coefficient as a function of  $k$ . (c) Mean silhouette width across multiple distance metrics as a function of  $k$ . These criteria yield inconsistent guidance, motivating the survival-supervised model-selection approach described in the main text.

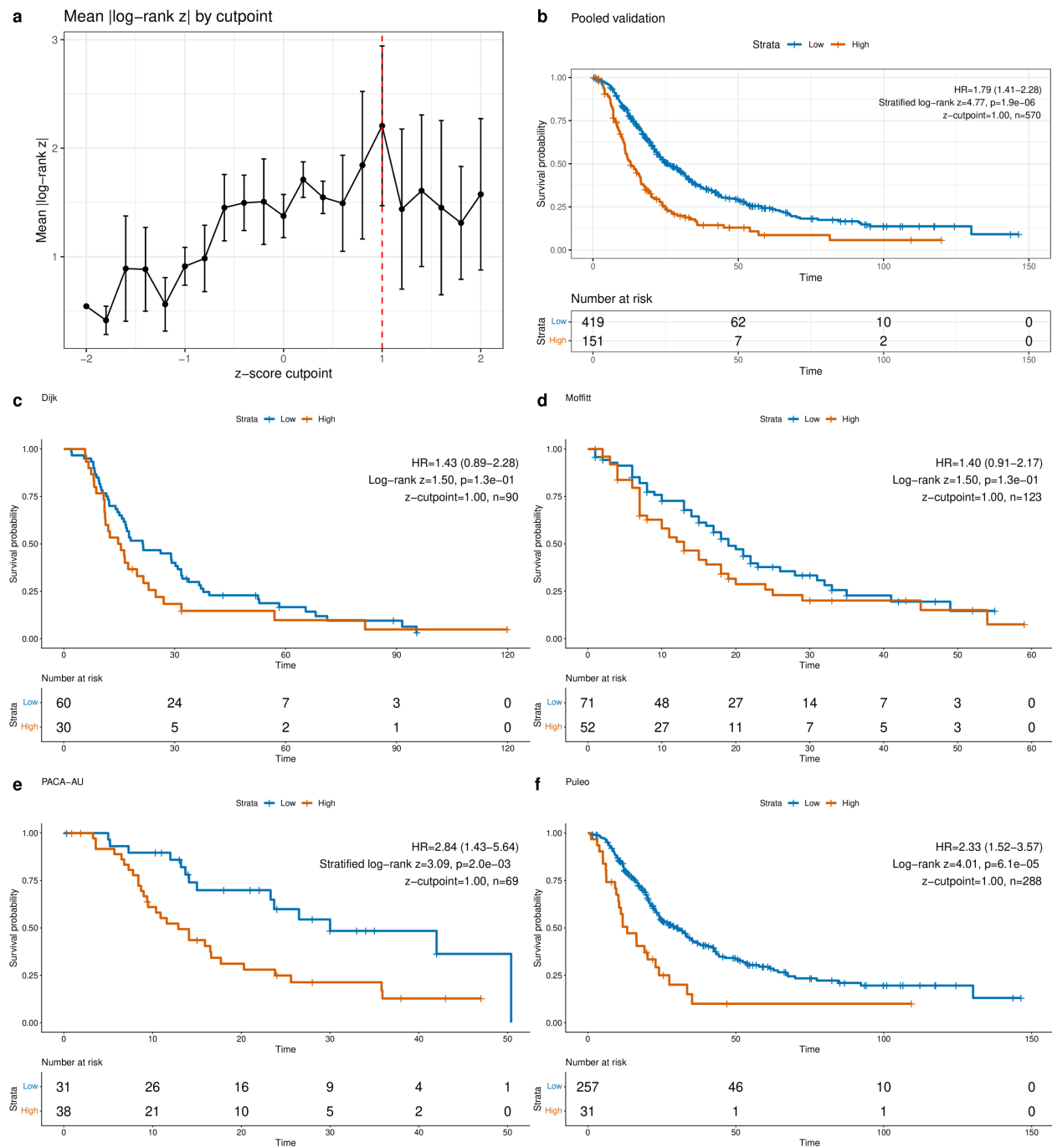


Figure S7: Cutpoint selection and external validation of dichotomized risk groups. (a) Cross-validated mean absolute log-rank z-statistic as a function of z-score cutpoint; red dashed line indicates the selected optimum. (b) Pooled validation Kaplan-Meier curves (log-rank test). (c-d) Per-cohort validation KM curves: (c) Dijk, (d) Moffitt. (e-f) Per-cohort validation KM curves: (e) PACA-AU (de-duplicated across array and RNA-seq platforms), (f) Puleo.

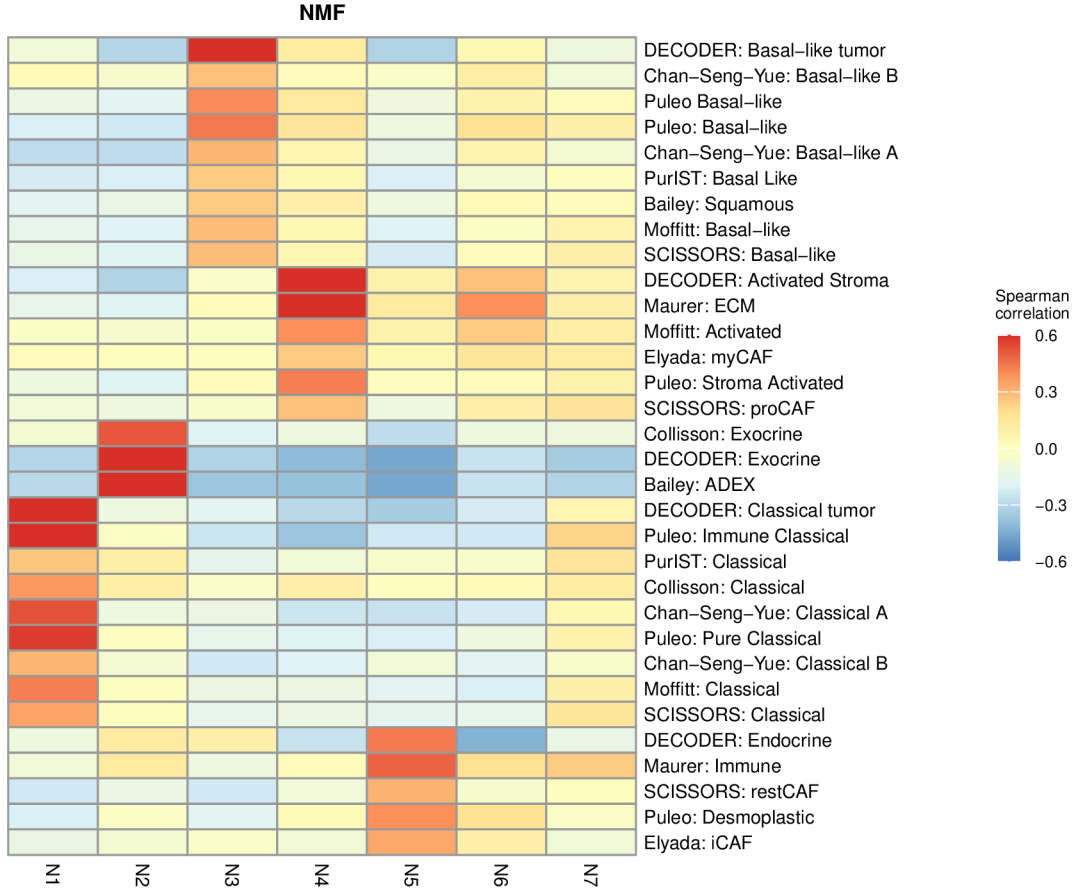


Figure S8: Gene-program correspondence for standard NMF at  $k = 7$  (BO-selected rank with  $\alpha = 0$ ), computed with the same Spearman-correlation analysis and reference programs as main-text Fig. 2a,b (top 50 genes per factor; reference programs with  $r > 0.2$  for at least one factor are shown). Compared with the matched-rank  $k = 3$  solution, the 7-factor solution shows increased fragmentation: multiple factors correlate with the same established PDAC signatures and no additional factor maps cleanly onto a distinct program.

Table S1: Summary of PDAC cohorts used in this study.  $N$  denotes the number of samples after all quality-control filters (histology, survival validity, dataset-specific exclusions). Patient-level metadata: OS = overall survival (time and event indicator); PurIST = tumor subtype classification (Rashid et al. 2020); DeCAF = CAF subtype classification (Peng et al. 2026). Expression data and molecular classifications for the validation cohorts were originally compiled for the DeCAF study (Peng et al. 2026). Events denotes the number of overall-survival events (deaths) with the corresponding event rate; outcome frequency was lower and more homogeneous in the training cohorts (50–52%) than across the validation cohorts (60–90%, pooled 67%), a case-mix difference relevant to the interpretation of external-validation performance.

| Cohort                                 | Platform   | $N$ | Events    | Role       | Accession       | Metadata          | Reference             |
|----------------------------------------|------------|-----|-----------|------------|-----------------|-------------------|-----------------------|
| TCGA-PAAD                              | RNA-seq    | 144 | 75 (52%)  | Training   | GDC (TCGA-PAAD) | OS, PurIST, DeCAF | Raphael et al. (2018) |
| CPTAC                                  | RNA-seq    | 129 | 64 (50%)  | Training   | GDC (CPTAC-3)   | OS, PurIST, DeCAF | Cao et al. (2021)     |
| Dijk                                   | RNA-seq    | 90  | 81 (90%)  | Validation | E-MTAB-6830     | OS, PurIST, DeCAF | Dijk et al. (2020)    |
| Moffitt                                | Microarray | 123 | 83 (67%)  | Validation | GSE71729        | OS, PurIST, DeCAF | Moffitt et al. (2015) |
| PACA-AU (array)                        | Microarray | 63  | 38 (60%)  | Validation | GSE36924        | OS, PurIST, DeCAF | Bailey et al. (2016)  |
| PACA-AU (RNA-seq)                      | RNA-seq    | 52  | 31 (60%)  | Validation | EGAS00001000154 | OS, PurIST, DeCAF | Bailey et al. (2016)  |
| Puleo                                  | Microarray | 288 | 181 (63%) | Validation | E-MTAB-6134     | OS, PurIST, DeCAF | Puleo et al. (2018)   |
| Training total                         |            | 273 | 139 (51%) |            |                 |                   |                       |
| Validation platform profiles total     |            | 616 | 414 (67%) |            |                 |                   |                       |
| Combined unique-patient validation set |            | 570 | 388 (68%) |            |                 |                   |                       |

PACA-AU contains 46 patients profiled on both the array and RNA-seq platforms; combined (pooled) analyses represent each patient once, retaining the RNA-seq profile for duplicated patients, consistent with the DeCAF preprocessing rule.

Platform-specific per-dataset analyses retain both records.

Table S2: Cross-method projection  $R^2$ : how reconstructable each factor’s gene-loading vector is from the other method’s full basis (OLS  $R^2$ ). DeSurv retains the bulk of the unsupervised structure (NMF N1 and N3 carry over at  $R^2 \geq 0.88$ ) and reorganizes it: the survival-aligned D1 program is essentially unreconstructable from the NMF basis ( $R^2 = 0.09$ ), i.e. not well reconstructed as a linear combination of the NMF factors, while the exocrine NMF factor N2 is poorly reconstructed from the DeSurv basis ( $R^2 = 0.38$ ).

| Factor                        | $R^2$ from other basis | Linear reconstructability       |
|-------------------------------|------------------------|---------------------------------|
| NMF N1 (tumor)                | 0.90                   | well reconstructed              |
| NMF N2 (exocrine)             | 0.38                   | poorly reconstructed            |
| NMF N3 (microenviron.)        | 0.88                   | well reconstructed              |
| DeSurv D1 (classical/restCAF) | 0.09                   | not linearly reconstructable    |
| DeSurv D2 (proCAF)            | 0.81                   | reconstructable ( $\approx$ N3) |
| DeSurv D3 (basal-like)        | 0.75                   | reconstructable ( $\approx$ N1) |

Table S3: Whole-matrix reconstruction  $R^2$  across the 1,970-gene analysis matrix for unsupervised NMF versus DeSurv at  $k = 3$ . Two denominators are shown. The centered  $R^2$  is the conventional definition, taken about the grand mean, and is the one to read as fraction of variation reconstructed. The uncentered  $R^2$  divides by  $\sum X^2$ ; because the analysis matrix holds within-sample ranks with grand mean  $p/2$ , the mean alone accounts for about 75% of that denominator, so uncentered values are high for any nonnegative low-rank fit and are reported only for comparability with the code. On the centered definition DeSurv gives up 6.0 percentage points of reconstruction relative to NMF (1.5 on the uncentered scale).

| Model              | Centered $R^2$ | Uncentered $R^2$ | Residual fraction (centered) |
|--------------------|----------------|------------------|------------------------------|
| NMF ( $k = 3$ )    | 0.812          | 0.953            | 0.188                        |
| DeSurv ( $k = 3$ ) | 0.752          | 0.938            | 0.248                        |

Table S4: Adjusted hazard ratios (per SD of the DeSurv linear predictor) for overall survival, before and after adjustment for standard clinical covariates and, where annotated, tumor purity, fitted separately in each cohort by complete-case analysis (per-cell  $n$  and events shown). The Covariates column lists the clinical variables entered (stage, grade, age, sex, nodal status, or resection margin, as available). After clinical adjustment the DeSurv score remains independently prognostic in Puleo, PACA-AU array and both training cohorts, and is attenuated to non-significance in PACA-AU RNA-seq, Dijk and Moffitt, the three smallest complete-case sets. Where tumor-purity annotation was available, purity adjustment left the estimate essentially unchanged wherever it could be measured on the full cohort (PACA-AU array 1.44 unadjusted versus 1.45 adjusted, PACA-AU RNA-seq 1.66 versus 1.65, TCGA 3.50 versus 3.60), which argues against the association being driven by tumor purity or composition. The exception, Moffitt, has purity for only 29 of its 123 patients. The smaller Dijk and Moffitt cohorts are individually underpowered, and their estimates are correspondingly unstable: Moffitt is non-significant both unadjusted and after adjustment, and its purity-adjusted model rests on only  $n = 29$  of its 123 patients. Training-cohort (TCGA, CPTAC) hazard ratios are in-sample.

| Cohort          | Covariates                    | Unadjusted                                      | Clinical-adjusted                               | Purity-adjusted                                |
|-----------------|-------------------------------|-------------------------------------------------|-------------------------------------------------|------------------------------------------------|
| PACA-AU array   | stage, grade, age, sex        | 1.44 (1.07–1.93), 0.016 ( $n=63$ , 38 ev)       | 1.58 (1.06–2.35), 0.025 ( $n=62$ , 37 ev)       | 1.45 (1.08–1.96), 0.014 ( $n=63$ , 38 ev)      |
| PACA-AU RNA-seq | stage, grade, age, sex        | 1.66 (1.15–2.38), 0.006 ( $n=52$ , 31 ev)       | 1.56 (0.95–2.55), 0.077 ( $n=51$ , 30 ev)       | 1.65 (1.15–2.36), 0.006 ( $n=52$ , 31 ev)      |
| Puleo           | sex, margin                   | 1.37 (1.18–1.59), $< 0.001$ ( $n=288$ , 181 ev) | 1.33 (1.14–1.55), $< 0.001$ ( $n=284$ , 177 ev) | —                                              |
| Dijk            | grade, age, sex, nodal        | 1.32 (1.06–1.66), 0.015 ( $n=90$ , 81 ev)       | 1.18 (0.92–1.50), 0.189 ( $n=85$ , 77 ev)       | —                                              |
| Moffitt         | sex, margin                   | 1.20 (0.98–1.48), 0.076 ( $n=123$ , 83 ev)      | 1.20 (0.94–1.53), 0.154 ( $n=98$ , 67 ev)       | 0.87 (0.50–1.54), 0.639 ( $n=29$ , 18 ev)      |
| TCGA (train)    | stage, grade, age, sex, nodal | 3.50 (2.63–4.66), $< 0.001$ ( $n=144$ , 75 ev)  | 3.32 (2.40–4.59), $< 0.001$ ( $n=142$ , 74 ev)  | 3.60 (2.68–4.84), $< 0.001$ ( $n=143$ , 74 ev) |
| CPTAC (train)   | stage, age, sex, nodal        | 3.72 (2.73–5.08), $< 0.001$ ( $n=129$ , 64 ev)  | 3.59 (2.59–4.97), $< 0.001$ ( $n=123$ , 61 ev)  | —                                              |

Table S5: External validation C-index by method and cohort. DeSurv at BO-selected rank ( $k = 3$ ) is compared with standard NMF at matched rank ( $k = 3$ ), elbow-selected rank ( $k = 5$ ), and BO-selected rank ( $k = 7, \alpha = 0$ ). Higher values indicate better discrimination. Note that the three unsupervised columns are not produced by one implementation:  $k = 3$  and  $k = 5$  are Lee multiplicative NMF (`NMF::nmf`), whereas  $k = 7$  is the DeSurv optimizer run at  $\alpha = 0$  with the same consensus initialization used for the supervised model. All columns are scored on the same union-of-top-genes support ( $n_{\text{top}} = 270$ ).

| Cohort            | DeSurv (K=3) | Std NMF (K=3) | Std NMF (K=5, elbow) | Std NMF (K=7, BO $\alpha=0$ ) |
|-------------------|--------------|---------------|----------------------|-------------------------------|
| Dijk              | 0.607        | 0.542         | 0.633                | 0.631                         |
| Moffitt_GEO_array | 0.562        | 0.521         | 0.552                | 0.578                         |
| PACA_AU_array     | 0.649        | 0.470         | 0.657                | 0.669                         |
| PACA_AU_seq       | 0.634        | 0.430         | 0.679                | 0.702                         |
| Puleo_array       | 0.636        | 0.534         | 0.642                | 0.651                         |

Table S6: Top 270 genes for DeSurv factor D1 (Factor D1 (classical tumor + restCAF-like stroma)), ranked by factor specificity score  $s_{ij}$ . The 270-gene cutoff corresponds to the Bayesian optimization–selected projection dimension ( $n_{\text{top}}$ ).

| Factor D1 (classical tumor + restCAF-like stroma) |            |          |         |          |          |
|---------------------------------------------------|------------|----------|---------|----------|----------|
| 1–45                                              | 46–90      | 91–135   | 136–180 | 181–225  | 226–270  |
| DMBT1                                             | FGFR2      | GOLGA8A  | PTPRN2  | RPL9     | PPP1R12B |
| IGFBP2                                            | COL9A2     | IL2RG    | FCGBP   | CD79A    | PIP5K1B  |
| S100A4                                            | COL22A1    | KIAA1211 | PLA2G2A | PROM1    | AHCYL2   |
| KRT23                                             | IRF7       | SFRP1    | FABP4   | SCD      | SST      |
| LIF                                               | SBNO2      | ADH1B    | EPPK1   | TMC5     | CYP3A5   |
| IFI6                                              | MGAT3      | MFSD2A   | CPXM2   | CCL14    | SMAP2    |
| PDLIM3                                            | SYNM       | CLU      | KIF1A   | PLXNB1   | ZG16B    |
| CLDN18                                            | APOD       | CLMN     | NGFR    | TMEM119  | SERPINF1 |
| MYO15B                                            | SEMA6A     | CHPF     | NFKBIA  | SORBS2   | CLDN3    |
| SYT8                                              | TNXB       | MTMR11   | SPOCK2  | C7       | NR1I2    |
| TFF2                                              | PTGIS      | TTR      | ATP2C2  | SLC12A7  | JUND     |
| TMPRSS2                                           | CCL18      | WFDC2    | NBEAL2  | QSOX1    | NCF2     |
| CHGB                                              | ZC3H12A    | C2orf72  | CAPN9   | UNC5CL   | CELSR1   |
| CST1                                              | CELSR3     | CCND2    | PTGDS   | LONRF2   | MUC17    |
| B3GNT7                                            | EMB        | ARHGAP4  | LSP1    | VSIG2    | INPP5D   |
| F5                                                | KCNJ15     | PKDCC    | LAMA5   | ADAMTSL2 | TM4SF5   |
| TNRC18                                            | SORBS1     | SRPX     | FASN    | ARHGEF16 | HNFB4A   |
| IGF2                                              | PAQR8      | CCL21    | GFRA1   | PCOLCE   | CFD      |
| ARSD                                              | SLC4A4     | CARD11   | PTPRH   | CFC1B    | HCLS1    |
| ZDHHC7                                            | ACSS1      | CTRB2    | FMOD    | SLC9A3   | SH3PXD2B |
| LENG8                                             | ST6GALNAC1 | GOLGA8B  | FHL1    | ACACB    | PCSK2    |
| ATP9A                                             | DES        | IYD      | PCDH20  | BCAS1    | HMGCS2   |
| TMPRSS3                                           | PRUNE2     | CORO1A   | CCR7    | RASD1    | FILIP1L  |
| CES1                                              | PLCB4      | SAA2     | CHIT1   | LGALS4   | ETNK1    |
| HOPX                                              | CADPS      | CDCA7    | PARM1   | MS4A1    | TNNI2    |
| CNN1                                              | FMO5       | BOC      | ITM2A   | CAPN6    | TOX3     |
| MIA                                               | RAP1GAP2   | LTB      | NAPSB   | SCCPDH   | FRZB     |
| CRIP2                                             | GALNT12    | COMP     | AGT     | PTPRCAP  | HABP2    |
| NEURL1B                                           | CCL2       | MANSC1   | FUT3    | VNN1     | GDA      |
| CHGA                                              | REG3A      | SLC19A3  | SIPA1L2 | PI16     | NCOR2    |
| ACTG2                                             | ZCCHC24    | STEAP2   | SPINK5  | SEMA7A   | DNAH2    |
| CXCL14                                            | LTF        | FOXA1    | ACSL1   | IL10RA   | DOK4     |
| TPPP3                                             | EFHD2      | WNK2     | BTNL8   | GALNT3   | HOXB9    |
| PIGR                                              | ATP11A     | SELL     | CORO2A  | GOLM1    | KIAA1324 |
| CYP27A1                                           | GP2        | PLEKHA6  | LAD1    | TNFAIP2  | TRIM2    |
| ADAM8                                             | HIPK2      | NFASC    | TMEM63A | CD79B    | FGFBP1   |
| FGFR3                                             | SLC26A9    | EGR3     | GC      | ENPP2    | IRX3     |
| IGF1R                                             | TACC2      | PLIN4    | C1orf21 | SQLE     | TINAGL1  |
| PLEKHB1                                           | TRAK1      | WDR72    | MIF     | KLF4     | SFRP4    |
| CPXM1                                             | SAA1       | AMY2A    | FOXA2   | PDLIM7   | APLNR    |
| ATP7B                                             | CLDN23     | IL16     | CCL19   | FOSB     | AGRN     |
| SLC12A2                                           | FOXJ1      | TJP3     | GSTA1   | RHPN2    | SLC25A25 |
| CCL20                                             | FRY        | SLC9A2   | LLGL2   | SCG2     | PODN     |
| AKNA                                              | GATA6      | LMOD1    | ACP5    | EMILIN1  | ITIH2    |
| COL16A1                                           | PLXNA2     | CPS1     | UGT2B15 | BACE2    | FLJ23867 |

Table S7: Top 270 genes for DeSurv factor D2 (Factor D2 (proCAF stroma)), ranked by factor specificity score  $s_{ij}$ .

| Factor D2 (proCAF stroma) |         |         |         |         |           |
|---------------------------|---------|---------|---------|---------|-----------|
| 1–45                      | 46–90   | 91–135  | 136–180 | 181–225 | 226–270   |
| UHMK1                     | RPS26   | SAMD9L  | ATP8A1  | IRF6    | RASSF2    |
| DDR2                      | MYLK    | DOCK2   | TGFBR1  | KLHL6   | ZNF83     |
| INHBA                     | COL11A1 | IGF1    | SYNPO2  | C1QTNF3 | CYTIP     |
| DYNC2H1                   | NFIB    | TACC1   | PDE4B   | PCLO    | SRPX2     |
| ZDHHC20                   | PTPN13  | CFH     | SGCD    | ITGA2   | LY75      |
| ITPR2                     | PDE3A   | DMD     | STK17B  | LRIG3   | HLA-DOA   |
| HMCN1                     | TRIM22  | PCDH7   | NUAK1   | SSPN    | STK38L    |
| LAMA2                     | DPYD    | MALAT1  | FCGR3A  | F2R     | RSAD2     |
| SLFN5                     | COL14A1 | NOTCH2  | GPNNB   | PDGFC   | HBB       |
| PLXNC1                    | CNTN1   | CDK6    | VCAM1   | CD109   | HERC2P2   |
| SVEP1                     | NCKAP1L | FBN1    | ADAMTS9 | CLIC4   | AOX1      |
| CYBB                      | PHLDB2  | SYNE2   | ITGB8   | SORL1   | PLIN2     |
| SYNE1                     | ITGA4   | RHOBTB3 | CR1     | AOAH    | PFKFB2    |
| SAMHD1                    | MACC1   | CALCRL  | SEPT11  | FMO2    | OGN       |
| SKIL                      | SEMA5A  | NRP1    | IFI44L  | RDX     | DOCK10    |
| ADAMTS12                  | MRC1    | LIFR    | ELL2    | IL1RAP  | RGS4      |
| BICC1                     | SLIT2   | STAB1   | GPR183  | GBP1    | AXL       |
| ITGA1                     | ZEB2    | FLRT2   | ZBTB16  | CXCR4   | FGD6      |
| COL8A1                    | ADAM10  | RNF144A | FRMD6   | WNK1    | CDC42BPA  |
| PDGFRA                    | SAMD9   | BCAT1   | PDE5A   | MFAP5   | SLC6A6    |
| ABCA1                     | MS4A7   | CILP    | CSF1R   | ALDH1A3 | RGS1      |
| TCF4                      | EFEMP1  | CCDC80  | PLEK    | CD53    | KIAA1324L |
| CEP170                    | BIRC3   | FAT4    | SLK     | OMD     | CPA3      |
| CD163                     | WISP1   | BNC2    | SLC7A2  | PLOD2   | CHL1      |
| MAN1A1                    | LAMA4   | LTBP1   | HEG1    | ITGAV   | SLC7A11   |
| MSR1                      | FAM198B | COL12A1 | GREM1   | MACF1   | GFPT2     |
| CCDC88A                   | PTPRC   | IL1R1   | VGLL3   | NID2    | RAI14     |
| F13A1                     | FAP     | SLA     | GBP5    | NAMPT   | LMAN1     |
| RPS28                     | ARRDC3  | MBOAT2  | SLC1A3  | ADAM28  | MECOM     |
| RALGAPA2                  | MYH10   | FNDC3B  | FGF7    | GNE     | FPR1      |
| ZEB1                      | FCGR2A  | PRRX1   | ABCA8   | ENAH    | EDEM3     |
| FBXO32                    | PARP14  | COL10A1 | ADAM12  | FAM129A | RASSF6    |
| INPP4B                    | DSE     | DDX60   | CYBRD1  | ECM2    | IQGAP2    |
| CDH11                     | EHF     | DOCK9   | TFRC    | VSIG4   | MIAT      |
| PKHD1                     | OSMR    | SRGN    | CTSK    | MTAP    | PRICKLE1  |
| AKAP2                     | ARID5B  | FOXO1   | COLEC12 | INMT    | CHST15    |
| CYP1B1                    | ASPN    | MS4A6A  | ALDH1L2 | SGMS2   | GEM       |
| CCL5                      | SLCO2B1 | ADAMTS2 | PGM2L1  | COL4A4  | CHRD1     |
| FGL2                      | FPR3    | CPVL    | FAM135A | CAPRIN2 | DSC2      |
| MAP1B                     | DOCK8   | CD93    | DOCK11  | CENPF   | RAB31     |
| PLXDC2                    | SLFN11  | DST     | EPHA3   | DCLK1   | SLC38A1   |
| LRRK2                     | ZNF532  | CDK14   | PAPPA   | EDNRA   | CNN3      |
| KIAA0754                  | SPON1   | MSRB3   | EGFR    | WNT5A   | FLT1      |
| EDIL3                     | SEMA3C  | RNF213  | RASA1   | GALNT5  | TMEM200A  |
| ITGBL1                    | FERMT2  | SEL1L   | AHR     | SLC38A2 | SLC2A3    |

Table S8: Top 270 genes for DeSurv factor D3 (Factor D3 (basal-like tumor)), ranked by factor specificity score  $s_{ij}$ .

| Factor D3 (basal-like tumor) |          |           |          |          |          |
|------------------------------|----------|-----------|----------|----------|----------|
| 1–45                         | 46–90    | 91–135    | 136–180  | 181–225  | 226–270  |
| HSPB1                        | HLA-H    | LAMB3     | MYL9     | IL1RN    | SDC1     |
| KRT7                         | HSPA1A   | FSCN1     | COL17A1  | TST      | KLK11    |
| MSLN                         | LGALS3   | TSPAN13   | EPS8L1   | RAPGEFL1 | PERP     |
| RPSAP58                      | FAM83H   | TSPAN8    | CDH3     | C15orf48 | ARHGAP23 |
| S100P                        | EPB41L1  | ALDH3B1   | RNASET2  | PFKP     | HPGD     |
| RPS16                        | EFNA1    | ANPEP     | TUBA1C   | STARD10  | AKR1C3   |
| IL32                         | TKT      | GSTP1     | RAP1GAP  | FOXQ1    | CRABP2   |
| SH3BGRL3                     | SLC2A1   | KLK10     | H19      | REG4     | GRB7     |
| IFITM2                       | PLOD1    | B4GALT1   | LTBP3    | GJB1     | SERPINA4 |
| RPS21                        | IER3     | GAS6      | AQP3     | SLC29A1  | SLC39A4  |
| SEMA4B                       | B3GNT3   | ABCC3     | MPZL2    | NOTCH3   | FA2H     |
| APOL1                        | PRSS3    | CLDN2     | ID1      | TRIM16   | RPL8     |
| S100A16                      | TOB1     | CEACAM6   | RARRES2  | TAP2     | MT1X     |
| DNAJB1                       | EGR1     | CREB3L1   | SLC25A6  | KRT19    | TCN1     |
| MT2A                         | PLA2G16  | MST1R     | RBP1     | CRP      | AKAP1    |
| TNFRSF21                     | OCIAD2   | CEBPB     | HSPA5    | SNHG5    | GJB3     |
| RPL28                        | LPCAT4   | TXN       | PER1     | KRT18    | DUOX2    |
| FXD3                         | BST2     | NR4A1     | ADAP1    | CA12     | HLA-DRB5 |
| SERPING1                     | EPHA2    | AP1M2     | TMC4     | MYEOV    | ARRDC2   |
| HMGA1                        | MUC5B    | SYNPO     | HSD17B2  | P4HB     | OAS1     |
| S100A14                      | KCNN4    | ANO1      | PTP4A1   | VILL     | ADRA2A   |
| HIF0                         | ITGA3    | SIK1      | PLAUR    | HKDC1    | NQO1     |
| KRT17                        | TSTA3    | EFNB2     | ANXA10   | STEAP3   | LRG1     |
| CSTB                         | IFI27    | CDKN1A    | PYGB     | PLEK2    | SERPINH1 |
| ASS1                         | MMP1     | S100A10   | CYP2S1   | CD68     | COL18A1  |
| EFNB1                        | SLPI     | PKP3      | ALDOA    | MAFF     | CERCAM   |
| MDK                          | CLTB     | MCM7      | CDCP1    | CD9      | CA9      |
| LY6E                         | SPINK1   | RARRES3   | IFITM3   | PLLP     | CDHR2    |
| CAMK2N1                      | JUP      | CEACAM1   | BAIAP2L1 | EPHB4    | ESRP1    |
| MGLL                         | PPP1R15A | ALDH2     | MGST1    | KRT16    | CRYZ     |
| PDZK1IP1                     | TRIM29   | CES2      | TFF1     | ABLIM3   | TAGLN2   |
| BCL2L1                       | PPP1R1B  | SLC6A8    | MUC4     | PTPRU    | HNFB1    |
| C19orf33                     | ARPC1A   | PPL       | MMP28    | MUC6     | BMP4     |
| SFN                          | GPRC5A   | NDRG1     | CLDN1    | FDFT1    | MT1E     |
| ITGB4                        | RAB25    | AZGP1     | TM4SF1   | GAPDH    | ANGPTL4  |
| TSPAN1                       | ZFP36L2  | S100A11   | TFF3     | EPS8L3   | CFB      |
| SLC16A3                      | CLDN7    | SEMA3B    | UNC13D   | CLDN10   | GDF15    |
| C4A                          | DDIT4    | TNFRSF12A | SERINC2  | ITPKC    | PADI1    |
| KLF5                         | FKBP4    | SOX9      | SPNS2    | LOXL2    | TMSB10   |
| IER2                         | LAPTM4B  | PMEPA1    | AQP5     | AMIGO2   | B2M      |
| ZFP36L1                      | MUC13    | CYR61     | IGFBP4   | DUOX2    | CAV2     |
| MAL2                         | PSCA     | EPCAM     | SLC7A5   | CLDN4    | CTSH     |
| SCNN1A                       | SDCBP2   | BHLHE40   | TESC     | CAPG     | PABPC1   |
| MALL                         | PROM2    | CX3CL1    | IFITM1   | CEACAM5  | KRT8     |
| BGN                          | LRRC8A   | SMAD3     | KCNK5    | VWA1     | TAP1     |

Table S9: Per-cohort overall-survival hazard ratios per SD in the three bulk treated PDAC cohorts.  $D_1$  marginal;  $D_2$  marginal and PurIST + DeCAF-adjusted;  $D_3$  marginal. Sample sizes: Linehan  $n = 66$  (22 events), Rash  $n = 85$  (73 events), Accept  $n = 70$  (57 events); basal-like prevalence Linehan 9% (6 cases), Rash 22%, Accept 16%.

| Cohort (setting)    | $D_1$ (marginal)        | $D_2$ (marginal)        | $D_2$ (+PurIST+DeCAF)     | $D_3$ (marginal)        |
|---------------------|-------------------------|-------------------------|---------------------------|-------------------------|
| Linehan (BR/LA)     | 0.58 (0.36–0.94), 0.027 | 0.60 (0.42–0.86), 0.005 | 0.46 (0.31–0.70), < 0.001 | 1.09 (0.68–1.77), 0.715 |
| Rash (metastatic)   | 0.81 (0.63–1.04), 0.096 | 0.87 (0.66–1.13), 0.283 | 0.77 (0.53–1.12), 0.178   | 1.09 (0.87–1.38), 0.455 |
| Accept (metastatic) | 1.10 (0.87–1.38), 0.427 | 0.78 (0.58–1.04), 0.091 | 0.73 (0.52–1.04), 0.078   | 1.11 (0.85–1.45), 0.445 |

Table S10: Arm-separated exploratory associations of the projected DeSurv programs (per SD) with clinical endpoints in the treated RASH/ACCEPT cohorts, analysed within each treatment arm. Overall survival and time to progression are Cox hazard ratios (HR); binary response (PR/SD vs PD) is a logistic odds ratio (OR); depth of response is a Spearman correlation ( $\rho$ ). Benjamini–Hochberg (BH) adjustment is across all unique arm-separated tests shown.

| Arm                   | Program | Endpoint               | $n$ (events) | Estimate (95% CI)   | $P$ ( $P_{BH}$ ) |
|-----------------------|---------|------------------------|--------------|---------------------|------------------|
| FOLFIRINOX            | D1      | Overall survival       | 16 (15)      | HR 0.88 (0.54–1.44) | 0.616 (0.834)    |
| FOLFIRINOX            | D1      | Time to progression    | 16 (13)      | HR 1.83 (0.90–3.71) | 0.094 (0.562)    |
| FOLFIRINOX            | D1      | Response (PR/SD vs PD) | 14 (10)      | OR 0.13 (0.01–1.51) | 0.104 (0.562)    |
| FOLFIRINOX            | D2      | Overall survival       | 16 (15)      | HR 0.80 (0.43–1.51) | 0.498 (0.775)    |
| FOLFIRINOX            | D2      | Time to progression    | 16 (13)      | HR 1.30 (0.70–2.39) | 0.404 (0.738)    |
| FOLFIRINOX            | D2      | Response (PR/SD vs PD) | 14 (10)      | OR 0.70 (0.23–2.09) | 0.525 (0.780)    |
| FOLFIRINOX            | D3      | Overall survival       | 16 (15)      | HR 1.99 (1.09–3.62) | 0.024 (0.504)    |
| FOLFIRINOX            | D3      | Time to progression    | 16 (13)      | HR 1.32 (0.71–2.46) | 0.378 (0.721)    |
| FOLFIRINOX            | D3      | Response (PR/SD vs PD) | 14 (10)      | OR 0.99 (0.30–3.19) | 0.981 (0.981)    |
| Gemcitabine           | D1      | Overall survival       | 24 (19)      | HR 1.01 (0.61–1.70) | 0.955 (0.979)    |
| Gemcitabine           | D1      | Time to progression    | 24 (15)      | HR 0.67 (0.34–1.31) | 0.240 (0.631)    |
| Gemcitabine           | D1      | Response (PR/SD vs PD) | 20 (12)      | OR 1.11 (0.38–3.21) | 0.846 (0.960)    |
| Gemcitabine           | D1      | Depth of response      | 20           | $\rho$ 0.06         | 0.802 (0.950)    |
| Gemcitabine           | D2      | Overall survival       | 24 (19)      | HR 0.54 (0.33–0.88) | 0.013 (0.504)    |
| Gemcitabine           | D2      | Time to progression    | 24 (15)      | HR 0.72 (0.44–1.18) | 0.198 (0.613)    |
| Gemcitabine           | D2      | Response (PR/SD vs PD) | 20 (12)      | OR 1.36 (0.59–3.17) | 0.471 (0.760)    |
| Gemcitabine           | D2      | Depth of response      | 20           | $\rho$ -0.03        | 0.904 (0.964)    |
| Gemcitabine           | D3      | Overall survival       | 24 (19)      | HR 1.25 (0.80–1.93) | 0.326 (0.721)    |
| Gemcitabine           | D3      | Time to progression    | 24 (15)      | HR 1.16 (0.69–1.98) | 0.572 (0.801)    |
| Gemcitabine           | D3      | Response (PR/SD vs PD) | 20 (12)      | OR 1.22 (0.53–2.83) | 0.645 (0.846)    |
| Gemcitabine           | D3      | Depth of response      | 20           | $\rho$ -0.15        | 0.538 (0.780)    |
| Gemcitabine/Erlotinib | D1      | Overall survival       | 48 (41)      | HR 0.79 (0.56–1.11) | 0.181 (0.613)    |
| Gemcitabine/Erlotinib | D1      | Time to progression    | 48 (40)      | HR 1.17 (0.84–1.61) | 0.357 (0.721)    |
| Gemcitabine/Erlotinib | D1      | Response (PR/SD vs PD) | 47 (33)      | OR 0.90 (0.49–1.66) | 0.731 (0.904)    |
| Gemcitabine/Erlotinib | D2      | Overall survival       | 48 (41)      | HR 0.81 (0.56–1.17) | 0.258 (0.637)    |
| Gemcitabine/Erlotinib | D2      | Time to progression    | 48 (40)      | HR 0.72 (0.50–1.04) | 0.081 (0.562)    |
| Gemcitabine/Erlotinib | D2      | Response (PR/SD vs PD) | 47 (33)      | OR 1.37 (0.68–2.77) | 0.374 (0.721)    |
| Gemcitabine/Erlotinib | D3      | Overall survival       | 48 (41)      | HR 0.78 (0.54–1.14) | 0.204 (0.613)    |
| Gemcitabine/Erlotinib | D3      | Time to progression    | 48 (40)      | HR 1.47 (0.93–2.33) | 0.101 (0.562)    |
| Gemcitabine/Erlotinib | D3      | Response (PR/SD vs PD) | 47 (33)      | OR 1.13 (0.57–2.22) | 0.731 (0.904)    |
| Gemcitabine/Afatinib  | D1      | Overall survival       | 46 (38)      | HR 1.12 (0.85–1.48) | 0.430 (0.750)    |
| Gemcitabine/Afatinib  | D1      | Time to progression    | 46 (33)      | HR 1.02 (0.74–1.40) | 0.913 (0.964)    |
| Gemcitabine/Afatinib  | D1      | Response (PR/SD vs PD) | 40 (31)      | OR 2.13 (0.98–4.65) | 0.057 (0.562)    |
| Gemcitabine/Afatinib  | D1      | Depth of response      | 40           | $\rho$ -0.12        | 0.446 (0.750)    |
| Gemcitabine/Afatinib  | D2      | Overall survival       | 46 (38)      | HR 0.98 (0.67–1.44) | 0.918 (0.964)    |
| Gemcitabine/Afatinib  | D2      | Time to progression    | 46 (33)      | HR 0.83 (0.55–1.24) | 0.354 (0.721)    |
| Gemcitabine/Afatinib  | D2      | Response (PR/SD vs PD) | 40 (31)      | OR 2.04 (0.78–5.37) | 0.147 (0.562)    |
| Gemcitabine/Afatinib  | D2      | Depth of response      | 40           | $\rho$ -0.24        | 0.142 (0.562)    |
| Gemcitabine/Afatinib  | D3      | Overall survival       | 46 (38)      | HR 1.04 (0.73–1.50) | 0.814 (0.950)    |
| Gemcitabine/Afatinib  | D3      | Time to progression    | 46 (33)      | HR 1.40 (0.92–2.15) | 0.119 (0.562)    |
| Gemcitabine/Afatinib  | D3      | Response (PR/SD vs PD) | 40 (31)      | OR 0.55 (0.20–1.48) | 0.235 (0.631)    |
| Gemcitabine/Afatinib  | D3      | Depth of response      | 40           | $\rho$ 0.24         | 0.141 (0.562)    |